

BUSINESS

Business is a activity aimed at earning profit. Business involves use of scarce resources and satisfaction of human needs, using such resources. Thus, business is an economic activity.

According to W.O. Wheeler,

“Business unit is a concern, company or enterprise which buys and sells, is owned by one person or group of persons and is managed under a specific set of operating policies”.

(CHARACTERISTICS OF BUSINESS

Business activity involves

1) Dealing in goods and services

Business activity deals with goods and services. Goods may be

- a) Consumer goods :- goods which are ready for direct consumption
- b) Capital goods :- goods which assist in the production of consumer goods.

Services refer to the facilities customers require such as banking, insurance, transportation, etc.

2) Risk and uncertainty

Risk refers to uncertainty of return. Profits are not guaranteed in business. Business units may suffer losses also.

3) Continuity

Business activity is characterised by continuity. Goods and services should be continuously traded. An isolated or a single transaction does not constitute business.

4) Satisfaction of human needs

The main aim of any business activity is to satisfy human needs. A profitable business unit is one which understands the needs of the customer and tries to satisfy them.

5) Profit motive

The incentive of earning profits. Keeps a person in a business and is also necessary for the continuity of the business. The objective of starting a business is to earn profit though he or she may suffer losses.

6) Transfer of title

Business activity is aimed at transfer of ownership of goods and services for value. In other words, the seller should transfer ownership of goods and services to the buyer.

7) Co-ordination of resources

Business activity requires resources, which are to be brought together. Lack of co-ordination of resources may not allow business to continue.

FACTORS INFLUENCING THE CHOICE OF SUITABLE FORM OF ORGANISATION

Before choosing a particular form of organisation the following points shall be taken into consideration.

1) Easy to form

The formalities to be fulfilled in order to start an organisation also have a great impact. A sole-trade business can be started without going through formalities. Whereas a joint stock company requires a lot of formalities to be followed, which is a time consuming and expensive affair.

2) Liability

In case of a joint stock company the liability of the share holder is limited to the value of shares held by them. Whereas in sole-trade business or partnership the liability is unlimited. Unlimited liability means that the private property of the owner can also be attached in case of Legal obligations.

3) Tax - Liability

Tax liability depends upon the earnings of a business. A business organisation Like a joint stock company needs to pay more tax when compared to a sole trader.

4) Government Regulations

while deciding about the form of organisation government rules and regulations are to be considered. A sole trade business does not require any rules and regulations to be fulfilled. Whereas a Joint stock company has to follow various regulations and rules to form and start its activity.

5) Ownership and Management

In sole trade and partnership type of business activity ownership and management are with same person (s). Whereas in the case of a company management and ownership are in two different hands.

SOLE - TRADE OR PROPRIETORSHIP

Sole trade is the simplest, oldest and the most natural form of a business organisation. Because the sole trader depends on his own resources, the business activity is carried-on a small-scale basis. Being the only owner and investor, a sole-trader has to bear all the risk and manage the business.

Definition

“A Sole-trader is a person who carries on business exclusively by and for himself. He is not only the owner of the capital of the under taking, but is usually the organiser and manager and takes all the profits or responsibility for Losses” JAMES STEPHENSON.

Characteristic of Sole Proprietorship

1) Individual initiative

Sole-proprietorship is started by the initiative of a single person. The ideas, resources, factors of production are co-ordinated by that person. Thus in the case of profits or Losses, the responsibility lies with the sole trader.

2) Easy to form

In majority of the cases, a sole-trader need not get a Licence from the government to start his business. Thus, formation of this type of business is very easy.

3) Limited Capital

Being the only investor, the capital that can be brought in by the sole-trader is limited. Thus, the business is run on a small-scale basis.

4) Unlimited Liability

The Liability of a sole-trader is unlimited. In case of Loss, the liability of the sole-trader is not limited only to his investment but his private property can also be attached to fulfill the business obligations.

5) Ownership and Management

The ownership and management of a sole-trade business is with the sole trader himself. It is for the sole-trader to plan and execute his ideas.

6) Limited area of operation

With the limited availability of resources, the area of operation of a sole-trade business is also very limited.

7) Motivation

In sole-proprietorship rewards and efforts are directly related. A sole-trader, being the sole person to enjoy profits, gets motivated and puts in extra effort to expand his business.

8) Secrecy

Business secrets of a sole-trader are very essential in order to avoid competitors from entering into same type of business.

Advantages of sole proprietorship

1. Easy formation :

Sole proprietorship is very easy to form as there are absolutely no legal formalities to be fulfilled to start it.

2) Personal touch with customers

The area of operation of a sole-trade business is small. Thus the number of customers are also less. So it is possible for the sole-trader to have personal contact with the customer and understand his tastes and preferences.

3) Prompt - decision making

Sole proprietor, being the only owner of the business need not consult anyone before

taking any decision regarding his business. Therefore, quick decisions can be taken by him.

4) Secrecy

It is easy to maintain secrecy in sole trade business, because there is no other individual who takes part in his business activity.

5) Beneficial to the society

Sole trade provides employment to people, who can invest small amount of capital. It is a form of self employment. And moreover, in a country Like India, where level of unemployment is high, sole-trade business is very much desirable.

6) Secrecy

It is easy to maintain secrecy in sole trade business, because there is no other individual who takes part in his business activity.

7) Better management and control

Management and control of sole trade is easy because the business is done on a small scale.

Disadvantages of sole proprietorship

1) Limited resources

A sole trader invests money into his business from his family sources. He cannot collect money/funds from the public like a Joint stock company. At the most he can borrow it from financial institutions.

2) Unlimited liability

The Liability of a sole trader is unlimited. His private property can also be attached for meeting business Losses.

3) No division of Labour

Sole trader being the only person handling the business, will not enjoy the benefit of dividing the work among specialised laboures. Any how, a sole trader can appoint people to help him, but the number of such people could be less taking into account his financial ability.

4) Uncertainty

A sole trade business will exist only till the existence of the sole trader. In case of his death, insanity, the business may cease to exist.

PARTNERSHIP

Partnership form of organisation is an improved version of sole proprietorship in many aspects. A sole trade business suffered from many drawbacks such as limited resources, lack of specialisation, uncertainty and so on. Many of these drawbacks are not a part of partnership form of organisation.

Definition

According to Indian partnership act, 1932, "partnership is relation between persons who have agreed to share the profit of the business carried on by all or any one of them acting for all"

Characteristics / Features of partnership

1) Association of two or more persons

In order to form a partnership form of organisation, there must be two or more persons. And these persons should be competent enough to enter into a partnership agreement.

2) Utmost good faith

As the number of owners of the business are two or more, it is very essential for them to trust each other. Otherwise suspicion and lack of faith among themselves will prove very fatal for the partnership.

3) Unlimited liability

As in the case of sole proprietorship, Liability of partner's in a partnership business is not limited to their investment. Their private property can also be accounted for, in case of any obligation.

4) **Restriction on transfer of share**

Without the consent of other partners, no partner can transfer his share in business to any outsider.

5) **Profit motive**

The underlining objective of partnership is to earn profits by engaging themselves into a legal business activity.

6) **Capital Contribution**

Every partner is required to bring into business an agreed amount of capital. But if the partners agree, any persons can be admitted into partnership without contributing any thing towards capital.

7) **Number of partners**

According to the Indian partnership act, the minimum number of partners should be two and the maximum number is 10 in case of banking business and 20 in case of non-banking business.

8) **Dissolution**

The closure of partnership is termed as "dissolution". When any of the partner dies, becomes insane or insolvent, the partnership is said to be dissolved.

9) **Principal and agent relationship**

In partnership form of organisation every partner acts as an agent and also the principal. Every partner may not be involved in the activities of the business. Thus one partner or partners act on behalf of others.

Advantages of partnership

1) **Easy to form**

A partnership form of organisation can be started with an agreement (written or oral) between the partners. No legal formalities are required.

2) **Availability of Large resources**

The resources of partnership are contributed by more than one person. Therefore the resources available are large, when compared to a sole-trade business.

3) **Better decision making**

The decisions taken by a partnership firm can be much better because the opinion of all the partners is taken into account. Unlike sole proprietorship where only one person takes a decision with the limited knowledge he has.

4) Sharing of risk

The burden of Loss, if any, is borne by all the partners, which makes partnership less risky, when compared to sole proprietorship.

5) Democratic management

All the partners, irrespective of their contribution to the capital of the organisation, will have equal right in the management of the business.

6) Flexibility

The operations of a partnership firm are flexible because there is no need for any prior permission from the government before making any change in the business activity, capital, etc.

Disadvantages of partnership

1) Unlimited Liability

As in the case of sole proprietorship, the liability of partners, in a partnership firm, is unlimited. It implies, that the private property of the partner's can also be assigned for obligations of business.

2) Limited resources

The resources of a partnership business are more when compared to sole proprietorship. But they are very less as compared to a joint stock company.

3) Instability

A partnership firm, can be dissolved at the death, insolvency or insanity of a partner. This makes it very instable.

4) Mutual distrust

The difference's between partners may become fatal for a partnership firm. Lack of confidence in each other may be the cause of these differences.

5) Lack of Public confidence

The accounts of partnership firm are not audited annually and published. This may lead to distrust among the public on the organisation.

6) Limitation on transfer of share

No partner has the right to transfer his share to any other party, without the consent of the other partners.

Partnership deed

Partnership deed or agreement is a document containing various aspects, which help in the smooth functioning of partnership and also easier settlement of disputes, arising at a future date.

An agreement can be written or oral. But, it is always better to have an written agreement, which will have a legal standing.

Contents of a partnership deed

Some of the important clauses of a partnership deed are

- 1) Name of the firm
- 2) Names and addresses of partners.
- 3) Nature of business.
- 4) Total amount of capital and contributions by each partner.
- 5) Profit sharing ratio among the partners.
- 6) Division of powers and duties among partners.
- 7) The method of valuation of goodwill.
- 8) Procedure for dissolution of the firm.
- 9) Procedure to be followed while admitting a new partner.
- 10) Rate of interest to be allowed on capital.

Kinds of partners

1) Active Partner

An active partner is one who takes active part in the day-to-day working of the business.

2) Sleeping or dormant partner

An sleeping partner does not take part in working of the business. But he contributes capital, shares profits and Losses.

3) Nominal Partner

A nominal partner does not bring in capital into the business, but only lends his name for name sake.

5) Secret partner

The membership of a secret partner is not disclosed to the public. But, a secret partner can take part in the working of the business.

6) Minor as a partner

A minor is a person who is below the age of 18 years. Minor can be admitted as a partner and can also be given a share in the profits of the business. But, he cannot be asked to bear the losses.

Registration of Partnership

Registration of a partnership firm is voluntary not compulsory. But an unregistered firm or its partners cannot go to court against third party in support of its claims.

A partnership firm, under the partnership act, 1932, can get itself registered by furnishing details like, name of the firm, details of location, addresses, details of partners, etc.

JOINT STOCK COMPANY

The draw backs of sole-proprietorship and partnership gave rise to company form organisation.

First Joint stock company was started in Italy in 13th century. In India the first company Act was passed in 1850. In 1956, a comprehensive act was passed, which is still in existence.

Definition

According to the companies act of 1956, "a company having a permanent paid-up nominal share capital of fixed amount divided into shares, also of fixed amount, or held as transferable as stock or divided and held partly in the one way and partly in the other, and formed on the principles of having for its members the holder of those shares or that stock and no other persons". is called as a Joint stock company.

According to Haney

"A joint stock company is a voluntary association of persons for profit whose capital is divided into transferable shares and ownership is required for its membership.

In simple terms, it is an artificial person created by law with perpetual succession and common seal.

Characteristics of a joint stock company

1) Artificial person

A company is an artificial person created by law. It has its own name and seal. It can perform all the activities of a businessman like purchase and sale of goods etc.

2) Limited Liability

The Liability of share holders is limited only to the face value of shares held by them. For example, if a person purchased 10 shares @ Rs. 100 each, his liability would be Rs. 1000 (10 shares x Rs. 100) only.

3) Management and ownership in separate hands

Shareholders are the owners of the business. But they do not manage the business. The board of directors appointed by them manage the company.

4) Transferability of shares

When in need of money, a shareholder can transfer his share in the company to other persons by following the procedure laid down for such a transfer.

5) Common Seal

Company, being an artificial person created by law, cannot sign. Therefore, the people who manage the company should affix the company's seal as an alternative to its signature.

6) Perpetual succession

Shareholders may change, but the existence of the company will continue without any interruption.

Kind of Companies

III. On the basis of ownership

- a) Government companies
- b) Holding companies
- c) Private companies
- d) Public companies

IV. On the basis of nationality

- a) Indian companies
- b) Foreign companies.

I. ACCORDING TO INCORPORATION

a) Chartered companies

Chartered companies are those which are incorporated under Royal charter issued by king or Head of State. According to this, special rights and privileges are granted to the company. The East India company, was incorporated under the same charter.

b) Statutory companies

The companies are formed according to a special act of parliament or state legislature.

The objects, rights, powers and duties of these companies are clearly listed out in the act, under which the companies are formed. Bank of India is an example of such company.

c) Registered Companies

These companies are formed under the companies act. Most companies in India are registered in India under the companies act of 1956.

III. ON THE BASIS OF OWNERSHIP

a) Government Companies

Government Company

According to the companies act of 1956, "Government Company means any company in which not less than 51% of the paid-up share capital is held by central government or by any state government or governments or partly by the central government and partly by one or more state governments and includes a company which subsidiary of government company".

b) Holding Companies

A holding company is the one which controls the composition of board of directors another company or holds more than half of the nominal value of the equity share capital in that company.

c) Private Companies

According to sec (3) of the Indian companies act, 1956, a private company is one in which

- i) restricts the right to transfer its shares, if any
- ii) Limits the number of its members to 50, excluding present and past employees.
- iii) Prohibits any invitation to the public to subscribe any shares or debentures.
- iv) Prohibits any invitation or acceptance of deposits from persons other than its members.

d) Public Companies

In simple terms a public company is one which is not a private company. Which means all the restrictions imposed on a private company does not apply to these companies.

IV. ON THE BASIS OF NATIONALITY

a) Indian Companies

Any company incorporated in India under the companies act of 1956, whether functioning in India or outside is called as an Indian Company.

b) Foreign / Multi - National Companies

A foreign company is one which is registered outside India, but operates in India through its branches or agents.

Merits of company form of organisation

1) Availability of Large resources

As the capital is collected from the public, by selling the shares in the primary market, a company enjoys the benefit of Large resources. When compared to sole proprietorship or partnership the resources available with a company are huge.

2) Permanent Existence

A company, being an artificial person, will continue to exist. Death of a shareholder or change in management will not wind-up the company.

3) Transferability of shares

The shares bought by a shareholder can be sold in the stock market. Whenever he needs money. It simply means the liquidity with shares is high.

4) Economies of Large scale production

With the availability of huge capital and management expertise, a joint stock company enjoys economies of marketing, production, specialisation, etc.

5) Diffused risk

Capital of a company is contributed by a large number of share holders, holding small number of shares. This spreads the risk among these large number people.

Demerits of company form of organisation

1) Difficulty in formation

In order to form / promote a company many Legal formalities are to be fulfilled. These formalities can be very expensive.

2) Delay in decision making

Decision making process in a company form of organisation will be very Lengthy. No single individual can take any decision. Only the board of director's can take decisions, which can be ratified by the shareholders in the annual general meeting.

3) Separate ownership and management

The management and the owners of the company are not directly related. Management unlike sole proprietorship and partnership, may not take personal interest in the growth of the company.

4) Lack of secrecy

When compared to other forms of business, a company form of organisation may not be able to maintain secrecy. Every matter has to be informed to the shareholders which may not remain secret thereafter.

5) Encourages Speculation

Speculation is the process of buying a share at a lower price expecting a higher price on its sale. This can be dangerous as shares can be manipulated and common investors trapped in such manipulation.

6) Evils of factory system

Evils of factory system like insanitation, air pollution, etc may become a problem for the society and also the employees working in the organisation.

What is a startup?

A startup is a newly established business, usually small, started by 1 or a group of individuals. What differentiates it from other new businesses is that a startup offers a new product or service that is not being given elsewhere in the same way. The keyword is **innovation**. The business either develops a new product/ service or redevelops a current product/service into something better.

What is Startup India?

Startups are becoming very popular in India. In order to develop Indian economy and attract talented entrepreneurs, the Government of India, under the leadership of PM Narendra Modi, has started and promoted Startup India initiative to recognize and promote startups

Startup company

A **startup** or **start-up** is a company or project begun by an **entrepreneur** to seek, **develop, and validate** a scalable **business model**. While **entrepreneurship** refers to all new businesses, including self-employment and businesses that never intend to become registered, startups refer to new businesses that intend to grow large beyond the solo founder. Startups face high uncertainty and have high rates of failure, but a minority of them do go on to be successful and influential.

How to register your startup with Startup India

Step 1: Incorporate your business

You must first incorporate your business as a [Private Limited Company](#) or a [Partnership firm](#) or a [Limited Liability Partnership](#)

You have to follow all the normal procedures for registration of any business like obtaining the certificate of Incorporation/[Partnership registration](#), PAN, and other required compliances.

Step 2: Register with Startup India

Then the business must be registered as a startup. The entire process is simple and online. All you need to do is log on to the [Startup India website](#) and fill up the form with details of your business and upload certain documents.

Step 3: Documents to be uploaded (in PDF format only)

a) A letter of recommendation/support

A letter of recommendation must be submitted along with the registration form. Any of the following will be valid-

(i) A recommendation (regarding innovative nature of business) from an Incubator established in a post-graduate college in India, in a format specified by the Department of Industrial Policy and Promotion (DIPP); OR

(ii) A patent filed and published in the Journal by the Indian Patent Office in areas affiliated with the nature of the business being promoted.

b) Incorporation/Registration Certificate

You need to upload the certificate of incorporation of your company/LLP (Registration Certificate in case of partnership)

c) Description of your business in brief

A brief description of the innovative nature of your products/services.

Step 4: Answer whether you would like to avail tax benefits

Startups are exempted from income tax for 3 years. But to avail these **benefits**, they must be certified by the Inter-Ministerial Board (IMB). Start-ups recognized by DIPP, Govt. of India can now directly avail IPR related benefits without requiring any additional certification from IMB.

Step 5: Finally, you must self-certify that you satisfy the following conditions

- a) You must register your new company as a Private Limited Company, Partnership firm or a Limited Liability Partnership
- b) Your business must be incorporated/registered in India, not before 5 years.
- c) Turnover must be less than 25 crores per year.
- d) Innovation is a must- the business must be working towards innovating something new or significantly improving the existing used technology.
- e) Your business must not be as a result of splitting up or reconstruction of an existing business.

Step 6: Immediately get recognition number

That's it! On applying you will immediately get a recognition number for your startup. The certificate of recognition will be issued after the examination of all your documents.

However, be careful while uploading the documents. If on subsequent verification, it is found to be obtained that the required document is not uploaded/wrong document uploaded or a forged document has been uploaded then you shall be liable to a fine of 50% of your paid-up capital of the startup with a minimum fine of Rs. 25,000

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Step 7: Other areas

a) Patents, trademarks and/or design registration

If you need a [patent](#) for your innovation or a [trademark](#) for your business, you can easily approach any from the list of facilitators issued by the government. You will need to bear only the statutory fees thus getting an 80% reduction in fees.

b) Funding

One of the key challenges faced by many startups has been accessing to finance. Due to lack of experience, security or existing cash flows, entrepreneurs fail to attract investors. Besides, the high-risk nature of startups, as a significant percentage fail to take-off, puts off many investors.

In order to provide funding support, Government has set up a fund with an initial corpus of INR 2,500 crore and a total corpus of INR 10,000 crore over a period 4 years (i.e. INR 2,500 crore per year). The Fund is in the nature of Fund of Funds, which means that it will not invest directly into Startups, but shall participate in the capital of SEBI registered Venture Funds.

Advantages:

1. Freedom of choice in all aspects of the business. Creativity can play a big part in starting and operating the business. The owner is in control of all aspects of the business, including the location and what the operation is to look like.

2. Low-cost entering: If you have limited capital available, you can start on a small scale.

3. Quality of Life: Entrepreneurs usually start businesses doing something they love. Growing a business from a hobby or part-time job gives entrepreneurs a chance to improve their lives because they are working in an environment important to them. Working in a positive business environment may not really seem like work to an entrepreneur.

4. Efficiency (Lean and Mean)

Established companies have high administrative overheads. Startups offer their services in a more efficient, cost-effective and competitive manner. They are likely to be aware of their limitations and tend to focus on their core strengths. This causes them to partner with other small organizations. Customers often benefit with a superior value proposition.

5. Flexibility

Your organization could have rather unique needs and demands for products and services. These might not be met by established service providers operating in a rigid manner. Startups are very flexible and are more likely to work at the hours, the place and in the manner that suits you.

Disadvantages

1. High commitment: You'll need to spend more time and effort to start up. This will include developing your customer base, establishing lines of credit and supply and finding experienced staff.

2. High risk- Success depends totally on you and your business talents. That's why future of your business remains uncertain and nothing can be taken for granted.

3. Delayed profitability- Where the market may not already be established, it may take longer to become profitable

4. More responsibility: All of the details of starting the business, including licenses, marketing, naming the business, finding product sources, etc. are the responsibility of the owner.

Internal & External Environmental Factors That Affect Business

Internal Environment Factors

Definition

The internal factors refer to anything within the company and under the control of the company no matter whether they are tangible or intangible. These factors after being figured out are grouped into the strengths and weaknesses of the company. If one element brings positive effects to the company, it is considered as strength.

On the other hand, if a factor prevents the development of the company, it is a weakness. Within the company, there are numerous criteria need to be taken into consideration.

Types

There are 12 types of internal environment factors:

1. Plans & Policies
2. Value Proposition
3. Human Resource
4. Financial and Marketing Resources
5. Corporate Image and brand equity
6. Plant/Machinery/Equipments (or you can say Physical assets)
7. Labour Management
8. Inter-personal Relationship with employees
9. Organizational structure or in some cases Code of Conduct
10. Quality and size of Infrastructure
11. Financial Forecast
12. The founders relationship and their decision making power.

External Environmental Factors

Definition

On the contrary to internal factors, external elements are affecting factors outside and under no control of the company. Considering the outside environment allows businessmen to take suitable adjustments to their marketing plan to make it more adaptable to the external environment.

There are numerous criteria considered as external elements. Among them, some of the most outstanding and important factors need to listed the are current economic situation, laws, surrounding infrastructure, and customer demands.

Types

Micro factors:

1. Customers
2. Input or Suppliers
3. Competitors
4. Marketing & Media

Macro factors:

1. Economic
2. Political/legal
3. Technology
4. Natural

7. a) Any 5 features of Sole trader, partnership and Joint stock company

Sole trader	Partnership
Unlimited liability	Profit and loss sharing
Privacy	Lawful business
Business secrecy	Principal-agent relationship
Decision making power	Number of partners
Easy to set up	Utmost good faith

unit-2 :

INTRODUCTION TO MANAGERIAL ECONOMICS AND ESTIMATION OF DEMAND.

The term Economics was derived from the two Greek words Oikos and Nomikos, it described economics as the science concerned with the problems of household management. An economic activity involves earning or spending of incomes . It mainly deals with activities of humanbeings in connection with satisfaction of their unlimited wants and limited resources with alternative uses . Thus the economic activities can be broadly divided in to ,

a) Production- to generates incomes .

b) Exchange- leading to trade and profit

c) Distribution- necessitates payment for the services rendered by factors of production in the form of rent , wages , interest and profits .

d) Consumption- involves spending income to satisfy wants .

MANAGERIAL ECONOMICS :

"Managerial economics is the use of economic models and modes of thought to analyse business situation".

- Mc Nair and Meriam

"Managerial economics is the integration of economic theory with business practice for the purpose of facilitating decision making and forward planning".

- Spencer and Siegleman

Components of Managerial economics includes both decision making and forward planning.

Decision making- involves choosing one course of action among several alternatives that are available. The concept of decision making arises at every stage of business viz., what to produce, where to produce, how to produce, for whom to produce, etc. Thus deciosion making is an important component of the business.

Forward planning- imply planning into the future . It is a known fact that future is always uncertain and unpredictable . It is an essential component and it underlined the fact that the managerial economist should apply the economic models developed by the pure economic theorist to find solutions to business problem .

NATURE OF MANAGERIAL ECONOMICS

1. Managerial economics is mostly micro in nature. It essentially relates to the decision making process of a firm or business unit, concerned with the internal objectives, management and administration of a business firm.
2. Though it is micro economic in nature, it has to focus on the area of macroeconomic environment that influence the decision making process like : nature of the economic system in which it operates; the policies and programmes pursued by the state(nation); economic legislations and functions of the land where the business operates; socio - cultural framework of the country.
3. Managerial economics is normative in nature i.e., it focusses on what it ought to be rather than what it is.
4. Managerial economics is prescriptive in nature.
5. It is both conceptual as well as metrical(quantitative).
6. Managerial economics is realistic in nature, it eliminates the assumptions of economic theories and applies the theories and models to reality.

SCOPE OF MANAGERIAL ECONOMICS:

The Scope Managing Economics Extends with The Following Areas:

1. Objectives of the firm.
2. Demand Analysis and Demand Forecasting
- 3.3. Production and cost analysis
4. Competition
5. Price - output decisions
6. Profit analysis
7. Capital budgeting
8. Marketing strategies

1.Objectives of the firm :

Managerial economics is mainly concerned with the functioning of a business firm. Managerial economics tries to analyse the firm's behaviour based on the objective it tries to realise .

2.Demand analysis and demand forecasting :

Demand analysis and demand forecasting is one main area of concern to the managerial economist . Demand analysis and forecasting in turn influences the production planning , product scheduling , pricing , cost control , inventory management,marketing strategies.

3. Production and cost analysis :

Managerial economics deals with the production and cost analysis . An understanding of the theories of production , cost concepts, cost functions and cost behaviour is an essential prerequisite for estimating profits, cost control and cost minimisation.

4. Competition :

A business firm should have a clear - cut idea of the market structure in which it is operating . The nature of competition influences its pricing and production policies . A comprehensive knowledge about the competitive conditions ultimately determines the business objective to be pursued by the firm .

5 . Price - output decisions :

Managerial economics is mostly concerned with pricing decisions,it has put forward several alternative methods of pricing such as full cost pricing , incremental cost pricing , target rate of return pricing etc. Business firms have to devise appropriate pricing method for the entire range of its products and the choice of appropriate pricing method is governed by the nature of the product, nature of competition, and the objectives of the firm.

6. Profit planning :

A business firm's ultimate long - term objective is to maximise profits . Hence profit planning and break - even analysis form an important component of Managerial economics .

7. Capital budgeting :

Capital budgeting and investment management have long - term repercussions on a business unit . Managerial efficiency in realizing long - term profits is mainly influenced by its ability to manage capital resources appropriately.

8. Marketing strategy:

Efficient production is one important concern of a business firm but more important is selling what ever is produced . This calls for devising relevant marketing strategies , deploying the sales personnel and unleashing the sales promotion campaign etc. Firms failing to sell cannot marketing becomes an important area of managerial concern .

There are several other areas that are part and parcel of the subject matter of Managerial economics .

Demand

Quantity of a good that a consumer wishes to purchase.

Definition: Demand is the quantity of a good that consumers are willing and able to purchase at various prices during a given period of time. The relationship between price and quantity demanded is an inverse relation i.e., they are inversely proportional to each other, with an increase in price the quantity demanded will decrease and vice - versa.

Factors influencing demand/ demand determinants:

Some of the common factors that influence the demand are:

1. The price:

The relationship between prices of a commodity and the quantity demanded is negative, means that an increase in price will induce a decrease in the quantity demanded and vice-versa. For example, if the price of a packet of milk were to rise from ₹25 to a price of ₹45, that would be a big price increase. Such increase causes the consumer to demand less of that product.

2.Price of related goods:

The principal related goods are complements and substitutes. A complementary good is a good that is used with the primary good. The relation between price and the quantity demanded for the complementary goods is negative.

Examples: automobiles and gasoline. (Perfect complements behave as a single good.) If the price of the complement goes up, the quantity demanded of the other good goes down. The other category of related goods are substitutes. Substitutes are goods that can be used in place of the primary good. The mathematical relationship between the price of the substitute and the demand for the substitute good is positive.

3. Personal Income:

In most cases, the more disposable income (income after tax and receipt of benefits) a person has, the more likely that person is to buy. With an increase in income level the consumer can be more likely to afford for the commodities he is willing. Hence, increase in income increases quantity demanded and vice-versa.

4. Tastes or preferences:

The greater the desire to own a good the more likely one is to buy the good. Demand is the willingness and ability to put one's desires into effect. It is assumed that tastes and preferences are relatively constant. Tastes and preferences of consumer will influence the quantity demanded for a product.

5. Consumer expectations about future prices, income and availability:

If a consumer believes that the price of the good will be higher in the future, he/she is more likely to purchase the good now. If the consumer expects that his/her income will be higher in the future, the consumer may buy the good now. This affects both price and demand.

6. Population:

If the population grows this means that demand will also increase.

7. The number of consumers in a market:

The market demand for a good is obtained by adding individual demands of the present, as well as future consumers of a good at various possible prices. The larger the consumer-base is for a good, the greater the market demand for it.

Demand schedule, demand function equation and demand curve:

Demand schedule:

Demand schedule shows the various quantities of particular product demanded at various price levels per unit of time in a table form.

Demand function:

The functional relationship between the quantity demanded and the factors affecting it is called as the demand function.

The demand function is expressed as

$$Q = f(P, Y, T, PR, \dots \text{Etc.})$$

Where,

P= price

Y= income

T= tastes and preferences of the consumer.

PR= prices of related goods (substitutes and complementaries).

Demand curve:

Demand curve is a graph depicting the relationship between the price of a certain commodity and the quantity of that commodity that is demanded at that price. Demand curves can be used either for the price-quantity relationship for an individual consumer (an individual demand curve), or for all consumers in a particular market (a market demand curve).

In economics the demand curve is the graphical representation of the relationship between the price and the quantity that consumers are willing to purchase. The curve shows how the price of a commodity or service changes as the quantity demanded increases. Every point on the curve is an amount of consumer demand and the corresponding market price.

The graph shows the law of demand, which states that people will buy less of something if the price goes up and vice versa.

Law of demand

The price of a product is a dominant factor. The law of demand states that all other things being equal an increase in the price is followed by a contraction of demand and a fall in price is followed by an expansion in the demand. In case of expansion and contraction of demand the consumer moves all along on a single demand curve. The terms expansion and contraction are associated with price changes while increase and decrease relate to changes in other factors other than the price .

Main features of the Law of Inverse Relationship or law of demand:

According to this law price and demand are inversely related .

Price is Independent variable and demand is dependent variable.

This law states the influence of change of prices on demand.

Assumptions of Law of Demand:

In order to prove influence of price on demand , this law assumes all other demand determinants are constant . . The above said feature of other things being equal implies that the law of demand is based on the assumption that all other demand determinants except price are constant . The following are the major assumptions of this law .

1. No change in income of consumer .
2. No change in price of other goods .
3. No change in tastes and preferences of consumer
4. No change in population
5. No change in weather conditions
6. No change in Govt.Policy
7. No anticipation of future prices .

Exceptions to the law of demand:

The law of demand fails to operate under the following conditions .

1)Giffen's paradox :

Sir Robert Giffen observed that the demand for inferior good like black bread expanded with an increase in price , as the low paid labourers in Britain were not in a position to substitute it with the expensive meat . This failed the law of demand .

2).Prestige Goods :

In case of status symbol products like diamonds the demand expands with rise in price and contracts with a fall in price . Thorstein Veblen found out this phenomenon hence it is named as Veblen effect .

3).Emergency Situations :

The law of demand does not apply in the period of emergency like war , curfew , flood , earthquake etc. During this period the accessibility of goods become uncertain . Therefore the consumers are ready to buy more quantity of goods even at higher prices.

4).Speculation Effect :

In the speculative market , it is observed that the demand for a u commodity is expanding with decrease in the price and contracting with price raise. Examples: stocks and shares, currency markets etc.

5). Ignorance :

Some time the quality of the product is measured by the consumer with profit the price of the product . If the commodity price increases they thought that the quality of the commodity improved and they tend to buy more .

ELASTICITY OF DEMAND INTRODUCTION:

The law of demand simply explains the direction in which the demand changes for a given change in price and nothing more .

Elasticity of demand explains the degree of correlation between demand determinants and quantity demanded i.e. whether demand is more responsive or less responsive to a change in determinants . Elasticity of demand measures the magnitude of change in the quantity demanded in relation to a change in price or income or any other variable affecting the demand.

Elasticity of demand can be defined as " the degree of responsiveness in quantity demand to a change price or any other demand determinant . There are four important types elasticity of demand .

Types of Elasticity of Demand 1. Price elasticity 2. Income elasticity 3 . Cross elasticity and 4. Promotional elasticity.

PRICE ELASTICITY OF DEMAND:

The concept of price elasticity of demand was brought in to the limelight by Marshal . Price elasticity of demand expresses the degree of correlation between price and quantity demanded . It can be used to measure whether the price and demand are correlated to a greater degree or lesser degree .

Price elasticity is measured as percentage change in quantity demanded divided by percentage change in price or proportionate change in quantity demanded divided by the proportionate change in price .

Types of Price Elasticity of Demand There are five varieties of price elasticity of demand .

1. Perfectly elastic demand

2. Perfectly inelastic demand
3. Relatively elastic demand
4. Relatively inelastic demand
5. Unitary elasticity of demand

1. Perfectly elastic demand : -

A small change in the price leading to an infinite change in the quantity demanded is known as perfectly elastic demand or infinite elasticity of demand . Numerically it is represented by the symbol ∞ . In this case the demand curve is parallel to the horizontal axis.

Perfectly elastic demand is having theoretical significance rather than practical importance . It is used only in economic theory , for example the demand curve faced by an individual firm under perfect competition is perfectly elastic . There is no other example of a commodity having perfectly elastic demand. But this can occur in case of a perfect competition that means for products which have many sellers and many buyers.

2. Perfectly inelastic demand : -

Irrespective of change in the price of a commodity if the demand remains constant or unchanged , is a case of perfectly inelastic demand . Numerically the elasticity of demand is equal to zero and the demand curve is perpendicular to the horizontal axis Salt is the best example of perfectly inelastic demand . The demand for salt remains unchanged whatever be the price .

3. Relatively elastic demand : -

The proportionate change in the quantity demanded is greater than the proportionate change in price is a case of relatively elastic demand . Numerically it is expressed as greater than unity i.e. > 1 . Normally comforts or non - essential goods will have elasticity greater than unity. In this case the demand curve will be more flat.

4. Relatively inelastic demand :

If the proportionate change in quantity demanded is less than the proportionate change in price then the elasticity of demand is relatively inelastic . Numerically it is expressed as less than unity i.e. < 1 . All the essential and necessary goods generally fall under the category of relatively inelastic demand and the demand curve will be more steeply sloping downwards .

5. Unitary elasticity of demand :

Proportionate change in quantity demanded is equal to 1 proportionate change in price is called unitary elasticity of demand . Numerically it is equal to one i.e., 1 . In this case the demand curve will be a rectangular hyperbola.

Importance or significance of price elasticity of demand:

1. Useful in formulation of taxation policy
2. Useful in decision making regarding public utilities, terms of trade.
3. Useful in devaluation
4. Used in monopoly pricing and price discrimination.
5. Useful in pricing of joint products.

INCOME ELASTICITY OF DEMAND:

The responsiveness in the quantity demanded to a change in the income of a consumer is known as income elasticity of demand . It can be measured as Proportionate change in the Q divided by Proportionate change in the income Y.

Income elasticity of demand is Positive in case of normal goods . In case of inferior goods it will be negative . The different kinds of income elasticity of demand are :

1. High - income elasticity :

In this case proportionate change in quantity demanded is greater than the proportionate change income . In case of high - income elasticity the demand curve will be more flat.

2 . Low income elasticity :

The responsiveness of demand being less than proportionate to the change in income is called low income elasticity , in which case the demand curve steeply slopes.

3. Income elasticity equal to unity :

It is a case where there is an equi - proportionate change in both the quantity demanded and in the income . The demand curve will be a 45° line through the origin.

4. Zero income elasticity :

If the responsiveness of demand to a change in income is zero , it is called as zero income elasticity of demand and the demand curve is perpendicular to X axis.

5. Negative income elasticity :

If the demand falls consequent upon a raise in the income.

Significance of Income Elasticity

Income elasticity is useful in decision - making and profit maximisation . For instance in times of boom or prosperity producers of luxury goods will gain due to high income elasticity while in times of recession they loose heavily . Conversely producers of necessary goods may not gain during boom but will have least adverse effect in times of depression due to low - income elasticity .

CROSS ELASTICITY OF DEMAND

Responsiveness in the demand for one commodity consequent upon the change in the price of another related commodity is called cross elasticity of demand . The related commodities are either substitutes or complementary goods . For complementary goods the cross elasticity will be negative and for Substitutes it will be positive .

Cross elasticity of demand can be measured as: proportionate change in quantity demanded Q of x divided by proportionate change in prices P of y.

Here,

Q of x - quantity demanded for commodity x

P of y - price of commodity y.

Types of cross elasticity:

1. Positive cross elasticity:

This can be seen in case of substitutes.

2. Negative cross elasticity:

Can be seen in case of complementaries.

3.unrelated cross elasticity:

This can be seen in case of unrelated goods.

Significance of cross elasticity

Useful in demarcating industry groups based on the cross elasticity of demand. For example all those firms whose products have a high positive cross elasticity can be demarcated under a single industry and products that have a negative or low cross elasticity can be brought under a different industry.

PROMOTIONAL ELASTICITY / ADVERTISEMENT ELASTICITY

Firms spend a large amount of money on advertisement, offering gifts, discounts etc. These expenses are often called as promotional expenditure. A large amount spent on advertisement may some times go baron or a small amount spent on advertisement may bring in disproportionate increase in sales.

Significance of promotional elasticity:

Promotional elasticity helps in understanding the responsiveness of demand in relation to advertisement expenditure. Promotional elasticity can be measured as $\text{Proportionate change in quantity demanded} \div \text{Proportionate change in advertisement Expenditure}$.

- **Social environment:** The social environment influences the demand. People living in posh localities create demand for cars and dish antennas. People in modest localities have high demand during festival season.
- **Government policy:** If the Government wants to encourage the consumption of certain commodities it offers tax concessions and subsidies and govt. wants to decrease the consumption it may increase the taxes to reduce the demand.

TYPES OF DEMAND

There are three types of demand namely price demand, Income demand and Cross demand.

• Price demand:

Price demand shows the functional relationship between price and quantity demanded at a particular point of time. In this case, the demand is treated as a function of price as shown below.

$$D = f(P)$$

D = demand f = function P = Price

Since price and quantity demanded are inversely related the price demand curve for all normal goods slopes downwards from left to right as shown in diagram-2.1

Inferior goods — Giffen goods:

However, it should be noted that in case of inferior commodities the price demand curve might slope upwards since the demand rises with a rise in the price and falls with a decline in the price as shown in diagram-2.3.

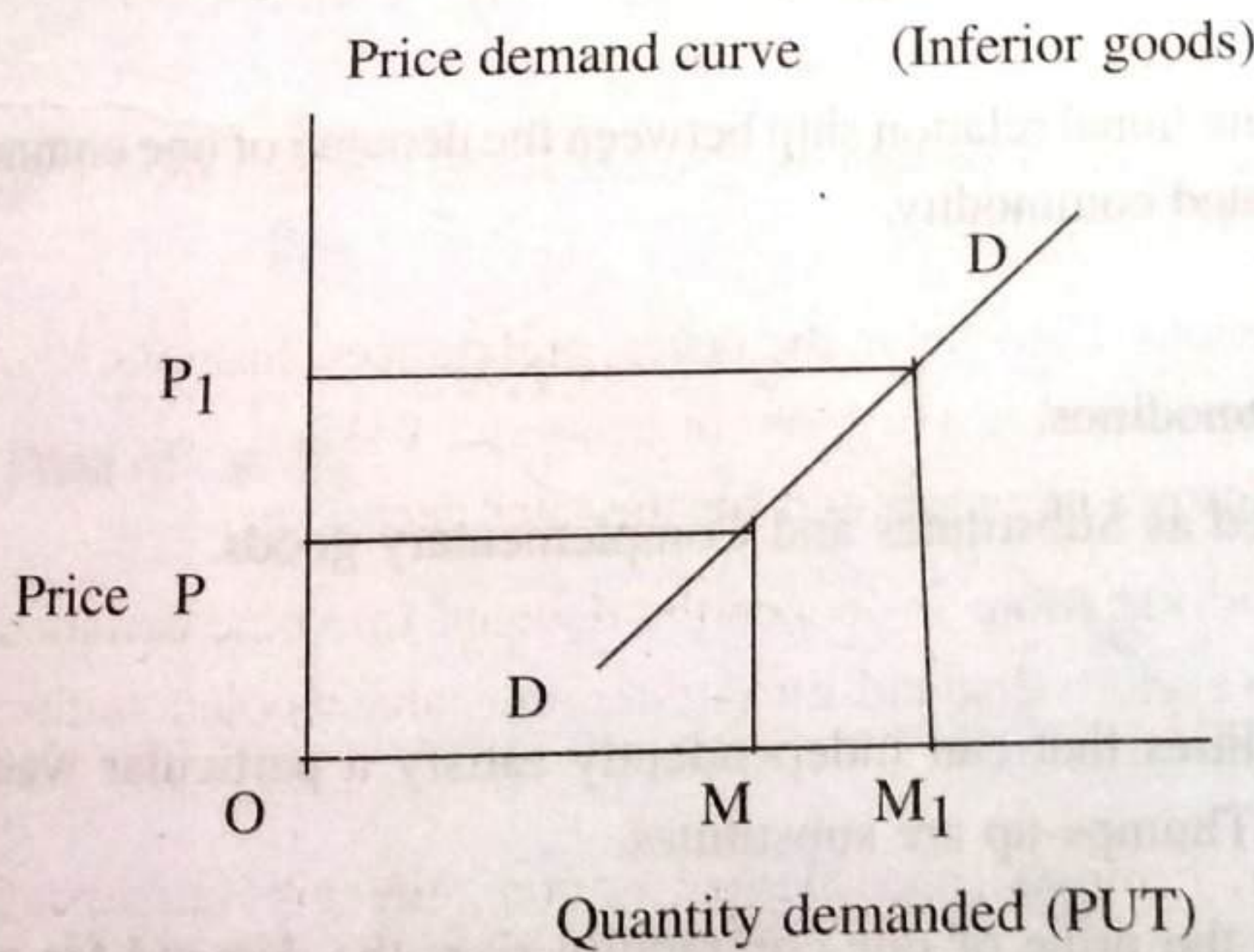


Figure 2.3

Breaks the fundamental law of demand: $P \propto Q$

Veblen goods — Conspicuous goods (quality premium goods)
 Giffen goods — non-luxury, low income goods, price rise — demand also rises

- **Income demand**

The functional relationship between the income and quantity demanded per unit of time is known as the income demand. Generally, the income and demand have positive correlation. In other words, higher the income larger will be the quantity demanded and vice-versa. The income demand curve therefore slopes upwards from left to right as shown in diagram-2.4.

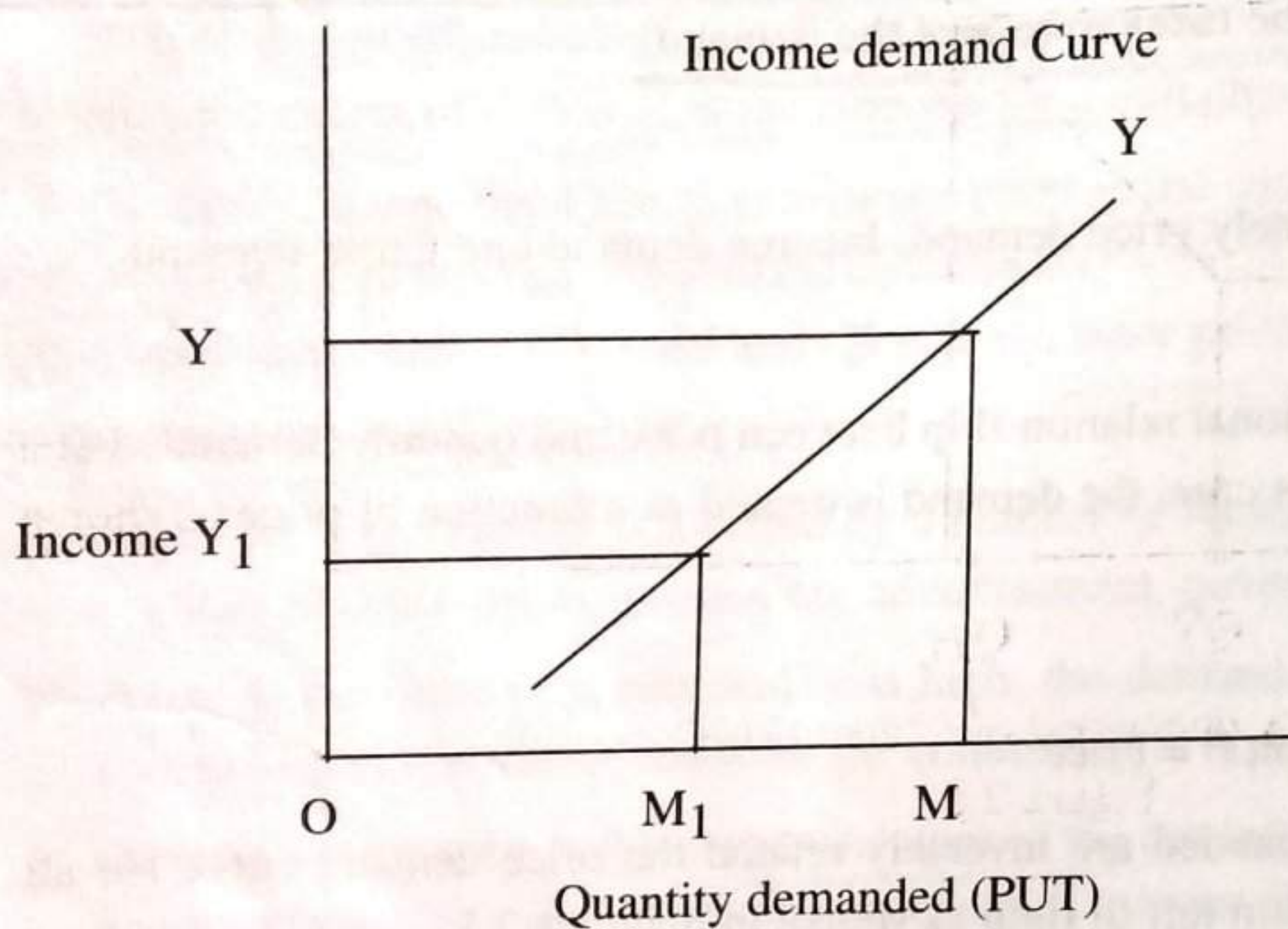


Figure-2.4

In diagram-2.4 at OY income the consumer demands OM quantity as the income falls to OY₁ the demand also falls to OM₁. Y_D is the income demand curve that slopes upwards from left to right.

- **Cross demand**

Cross demand expresses the functional relation ship between the demand of one commodity and the prices for another related commodity.

Where $D_X = f(P_Y)$

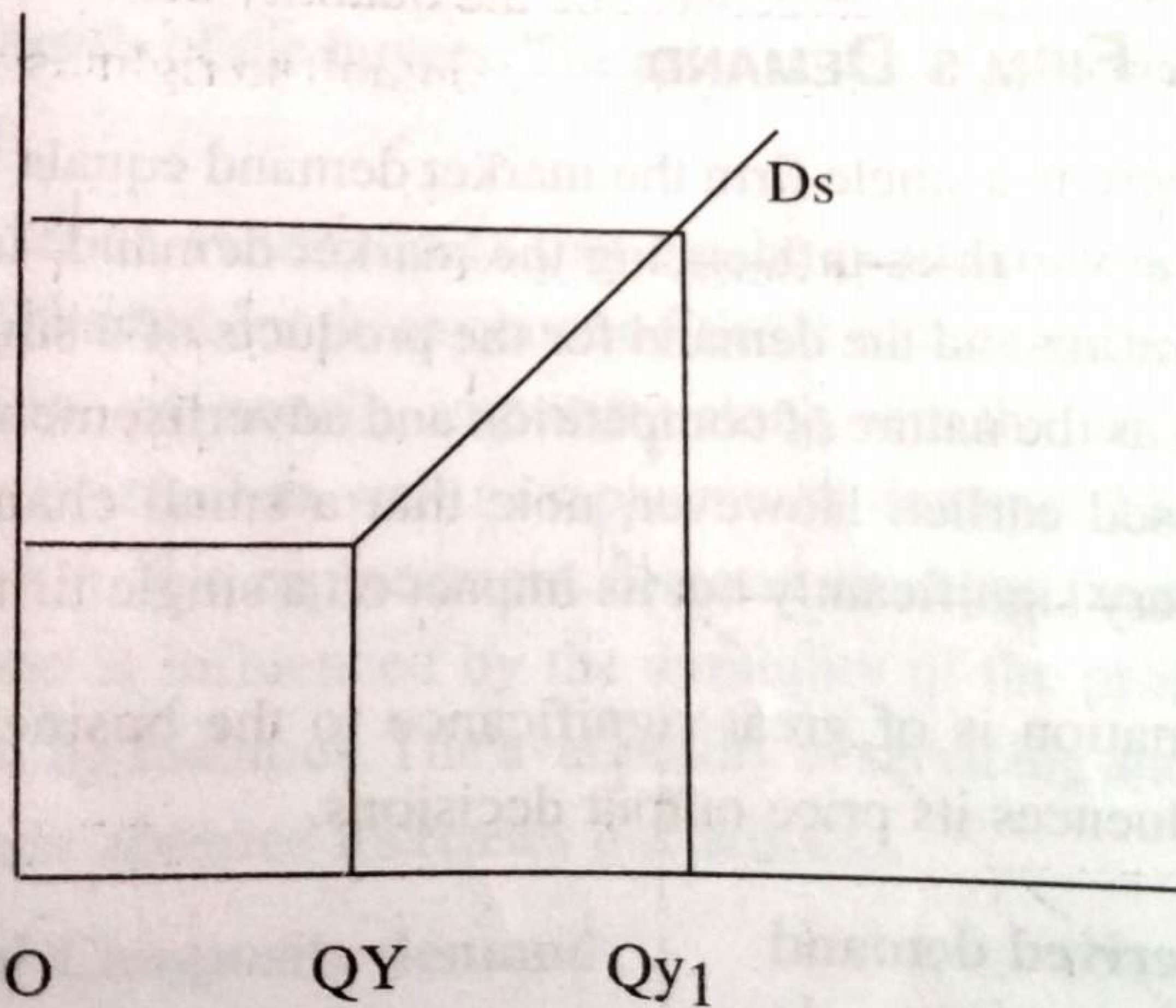
Here X and Y are related commodities.

The related goods are classified as Substitutes and Complementary goods.

Demand Theory and Analysis

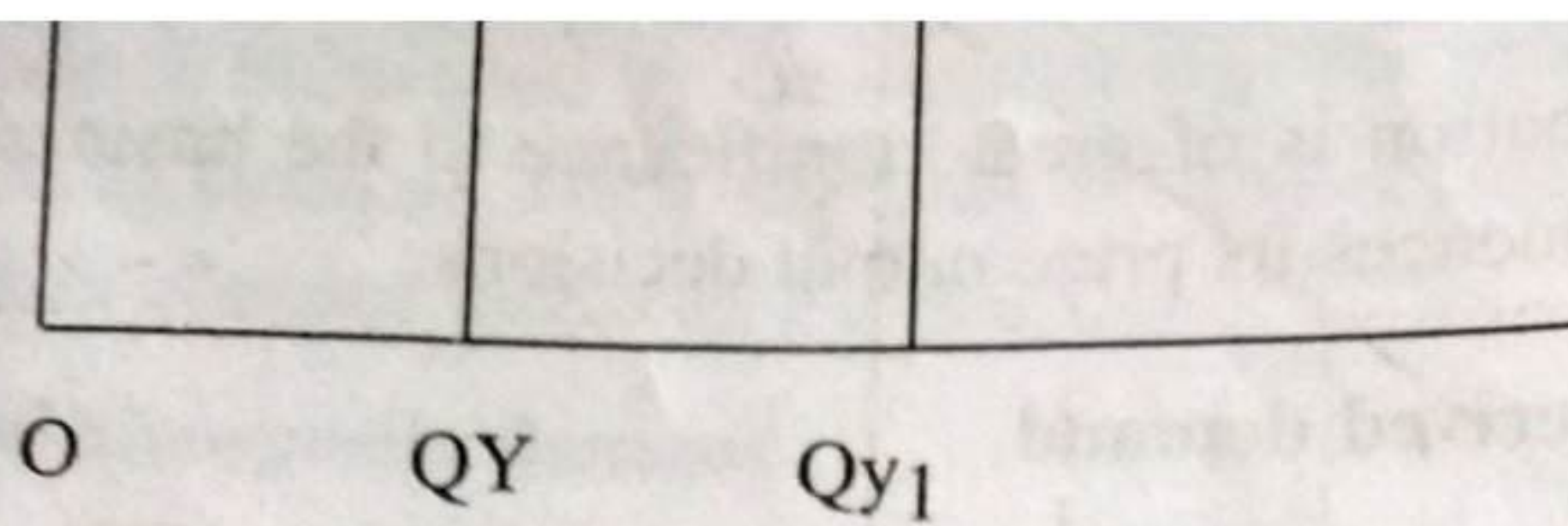
diagram 2.5. When the price of Tea increased from OP to OP_1 the demand curve shifts from QY to QY_1 . DS is the demand curve for substitutes that is sloping

Demand curve for substitutes



Quantity demanded of Coffee (PUT)

Figure-2.5



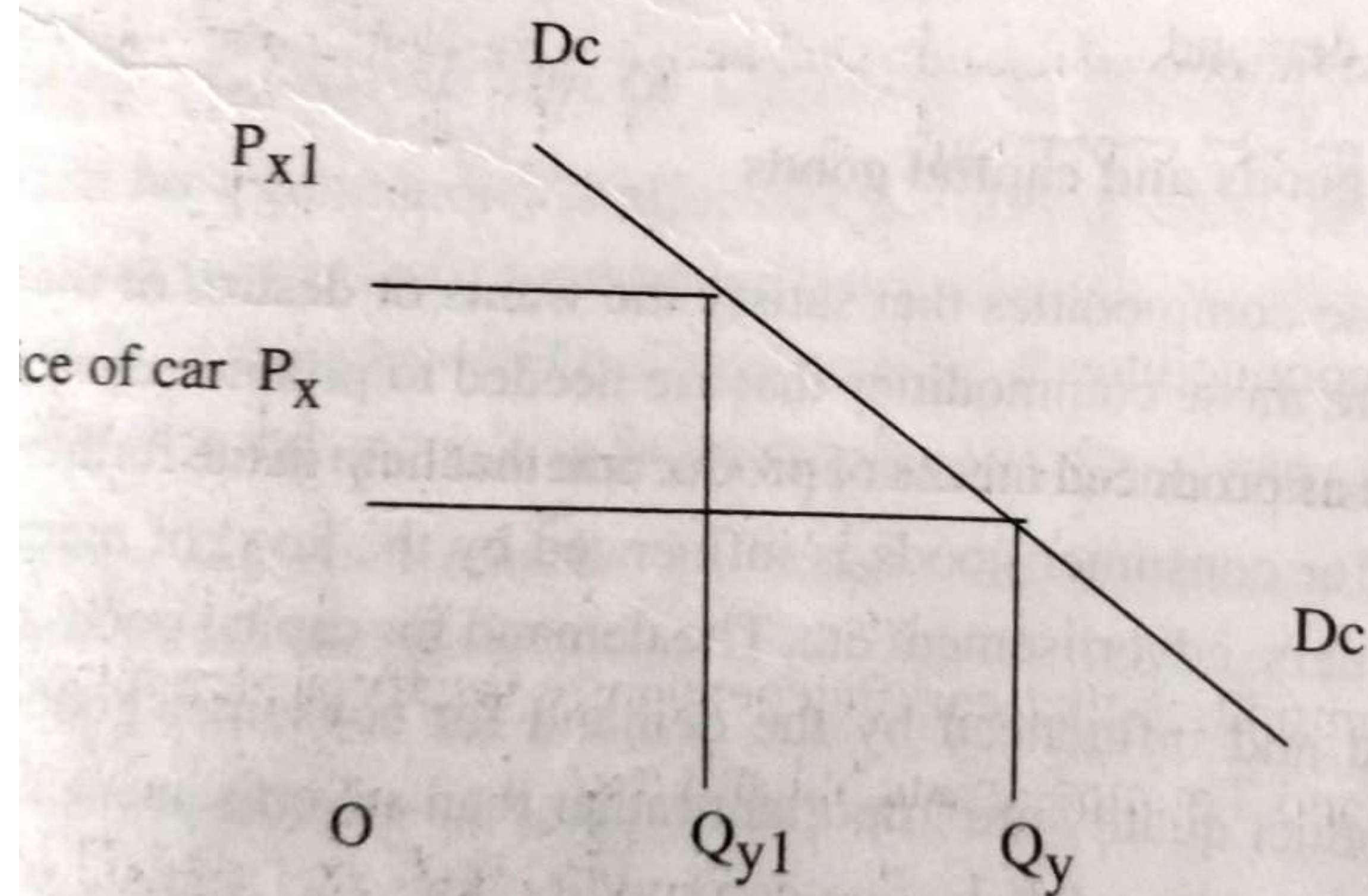
Quantity demanded of Coffee (PUT)

Figure-2.5

ntaries:

Complementaries are those goods, which are required together to satisfy one particular want. For example, car and petrol. For complementary goods, the price and quantity demanded are interrelated. An increase in the price of petrol reduces the demand for car. Therefore, the demand curve slopes downwards from left to right as shown in diagram 2.6.

Demand curve for complementary goods



Quantity demanded of petrol (PUT)

Figure-2.6

DEMAND FORECASTING

Demand concepts and demand theories are theoretical in nature . Price elasticity , income elasticity and cross elasticity of demand throw light on how micro and macro economic trends affect the demand for a commodity .

These tools will be applied only when proper data is available . In reality a business manager has to operate under conditions of uncertainty and should take decisions relating to pricing and production. Several methods of demand forecasting are available to the business decision makers, from which they have to choose the appropriate method suitable to the product and the purpose .

Importance:

Demand forecasting is important for production planning, production scheduling, inventory management and so on.

METHODS OF DEMAND FORECASTING

The methods of demand forecasting are broadly grouped under two heads called opinion survey methods and statistical methods .

The opinion survey methods are again classified as

- 1 . Consumer's opinion survey
- 2 . Expert opinion survey
- 3 . Collective opinion survey
- 4 . Test marketing or controlled experimentation method
- 5 . End use method

The statistical methods of demand forecasting include

- 1 . Trend projection
- 2 . Regression technique
- 3 . Moving average method
- 4 . Leading indicators method

Survey methods:

1. Expert opinion survey:

Under this method the experts are requested to provide their opinion about the future demand of a product. Generally sales representatives and sales officers in a company acts as experts to assess the demand for the products at different locations like regions, cities, etc.,)

This method is simple and inexpensive. Promotes cooperation among sales executives and the results will be accurate and reliable. Sales officers are those who have experience and knowledge regarding the market as they frequently get in touch with the market and consumers, hence their opinion survey forecasts the demand for a product.

2. Market experiment method:

This can also be called as consumer's survey method because market is a pool of consumers. This method carries out the studies and experiments on consumer behaviour under actual market conditions on the basis of population, income level, cultural background, tastes and preferences of consumers. The market experiments are carried out with the help of changing prices and expenditure, so the resultant change in demand are recorded for forecasting the demand. This method is expensive and not suitable for small scale organization's.

3. Delphi- technique:

This technique is based on the results of multiple rounds of questionnaires sent to a panel of experts. The questions are asked from a group of experts to obtain their opinions on demand for products in future. The forecasts obtained are cross checked among experts to reach more accurate decision making.

This method is both cost effective and time effective.

4. Test marketing or controlled experimentation:

Firms resort to test marketing when they launch a new product or likely to change the design or model of the existing product. This method is also popularly known as controlled experimentation method as the product will be launched in a segmented market like a city/town, a region, to identify its demand potential. Secondly test marketing should be continued till the repurchase cycle commences. Test marketing can be under taken in a single market or different markets with different features of the product. Test marketing will no doubt be a useful method but at the same time it is an expensive proposition.

Example: Test marketing of Lipton iced tea by HUL at Hyderabad City.

5 . End use or input – output method :

The end use method is applied for forecasting the demand for input products which are used in the manufacture of some other final goods. The demand for the final product can be an indicator for the demand for intermediate products. And this method focuses and identifies the higher consumption and lower consumption of a product or services to assess the demand.

Statistical Methods:

1. Trend projection method:

When time series data is available relating to the sales of a product, the demand can be forecasted through the trend projection method. It should be remembered that the time series data of sales includes cyclical changes in sales (C), seasonal

variations (S), trend (T) and irregular movements (I) .

Sales = (T , C , S , I)

Trend shows the long time tendency of the data . It is the result of changes in population , level of income , competitive conditions , changes in capital technology etc.

Cyclical changes in sales arise due to the periodical swings in the level of economic activity such as prosperity or recession operating in the economy . These changes are longer in duration and studying these variables is necessary for identifying the turning points .

Seasonal variations in sales occur at particular points of time in a year . The seasonal changes may take the form of climatic variations , festival seasons , holidays , marriage season etc.

Irregular variations cannot be predicted hence are not measurable like an earthquake , cyclone , war , or an epidemic breaks out.

The time series data can be used to forecast the sales by fitting a trend projection line to the observed data .

Free hand method of trend projection :

This is the easiest way of forecasting the demand . In this method time series data will be plotted on a diagram and a free hand trend line will be fitted in such a way that it passes through the closest points on the scatter diagram .

2. Regression technique or method of least squares :

The regression technique is also known as the method of best fit . In this case sales forecasts are made assuming 1) that there is single determining variable that is sales are a function of time and 2) it is also assumed that there is a linear relationship between

the variables . In this method the forecast can be made using the equation $Y = a + bx$ Where $y = \text{sales}$, $a =$ The y axis intercept i.e., constant , $b =$ The coefficient of the determining variable x , and $x = \text{time}$.

3. Moving Averages Method:

This is the method of demand forecasting used for those products whose demand fluctuates mostly due to seasonal disturbances . Moving Average Method is also known as smoothing method as it tries to eliminate the extremities by averages . When sales data is available continuously for a long period of time , the moving average method will be applied . While applying moving averages one can take odd or even number of periods (months or years) . Generally odd number of periods is preferable to even number of periods for calculating moving averages.

4. Leading indicators method :

If proper time series data is not available and the available data does not show particular trend, the forecaster has to rely upon secondary sources of data for forecasting. Sometimes changes in demand for one product occurs before an increase in the demand for another then the demand for the first acts as an indicator for the likely demand for the second. As an example increased availability of bank loans acts as indicator for the likely increase in the demand for commodities like steel and cement.

INTRODUCTION

Commodities in their natural state do not possess any utility unless they are given a particular shape or form. Putting utility into matter is called Production. Production is the result of the collective efforts of the four factors of production land labour capital and organization

Production function: Production function explains the functional relationship between the inputs and output. To get a certain amount of output what combination of inputs is required will be given out by the production function. For example

$$Q = f(K, N, L \dots \dots E)$$

Where

- Q = output
- f = Function
- K = Capital
- N = Labour
- L = Land
- E = other inputs

Assumptions of production function : The production function is based on the following important assumptions.

1. The level of technology remains constant.
2. The production function relates to a particular period of time.
3. The factors of production or divisible into small on.
4. The firm uses the best and most efficient technique

MANAGERIAL USE OF PRODUCTION FUNCTION

The production function is very much helpful to managers. The following are the managerial uses of production function.

- a) Helpful in determining least cost combination.
- b) It helps to determine optimum level of output.
- c) Useful in drafting plans.
- d) Supports in taking cost control decisions.

function using logs.

ECONOMIES OF LARGE SCALE PRODUCTION

The economies of large-scale production are classified as internal and external economies.

Internal economies and diseconomies: These are the advantages or benefits, which an individual firm obtains when its size expands. They are internal to an individual firm. The internal economies are classified as 1) Technical 2) Managerial 3) Commercial 4) Financial and 5) Risk-bearing economies.

- 1) **Technical economies** arise out of the use of modern and improved techniques of production. It is possible that large firms can make use of the latest technology that may either increase the production with the same investment or conversely it may enable the firm to produce the same level of output at a lesser cost. These economies arise out of dimensional changes of using bigger machines. For example a double decker bus carries twice the capacity of passengers but there will be a lot of savings in the form of single engine, fuel efficiency, wages of a driver and conductor and so on. Similarly when the surface area of a vessel is doubled the cost will increase only by 60% i.e. 6/10 rule applies. It is these technical economies that give rise to either increased production or reduced cost. Further large firms can make optimal utilization of indivisible factors of production i.e. men and machines leading to increasing returns or decreasing costs

- 2) **Managerial economies.** These are the economies arising out of division of labour and improved managerial abilities. Large firms can introduce the complex division of labour and derive all the benefits accruing out of it. Similarly they can deploy right man for the right job and achieve a high degree of managerial efficiency by departmentalizing the work.
- 3) **Commercial economies or marketing economies.** These are the benefits that large firms get from bulk buying and bulk selling. These firms can get their materials at reasonable rates directly from the manufacturers and avoid a number of intermediaries. Similarly they can sell their products through their retail outlets and derive the best prices. These firms can also appoint expert sales personnel to market their produce and incur advertisement expenses on a large scale.
- 4) **Financial economies.** Large firm can obtain finances very easily at reasonable terms. Financial institutions prefer large firms for lending as they can get good collateral security. Large firms can also mobilize financial resources for fixed capital needs by way of public issue of shares and debentures. Timely and adequate availability of funds minimizes the cost of financing projects.
- 5) **Risk-bearing ability.** Large manufacturing units have the ability to reduce or avert the risks easily and quickly. If they sense that there is a likely fall in the demand for their products they can take remedial action by improving quality or design change or by product diversification.

EXTERNAL ECONOMICS AND DISECONOMICS

External economies arise when an industry expands either with the entry of new firms or expansion of size by the existing firms. The benefits of these economies accrue not to one firm but to all the firms operating in the same industry. These external economies are classified as

- 1) Economies of concentration
 - 2) Economies of information
 - 3) Economies of disintegration or specialization
- 1. Economies of concentration:** when a cluster of firms producing identical product get concentrated at a particular place a number of supporting services will come into existence. For example infrastructure facilities such as power water transport and communication network will be created there. Educational institutions imparting the skills suitable to that industry will develop¹. It is easy to get the laborers with requisite skills in that location. Banking and financial institutions specialized in financing a particular line of activity also comes into existence. All these advantages arising out of concentration of industry are available to each one of the firms and help them in increasing productivity or in the alternative in reducing their unit costs.

2. Economies of information: A group of firms when localized can form themselves into an association of manufacturers². Collectively they can publish trade-journals and also establish Research and Development centers. They can undertake market surveys about the demand for their products. Through Research and development they can develop new techniques or products or processes. The entire information will be disseminated to all the firms through the trade journals. All the firms in the industry could reap the benefits of this information.

3. Economies of Specialization: when an industry gets localized the firms instead of producing the entire product may disintegrate the work into different processes and specialize in one process. This specialization brings in all the advantages associated with division of labour and internal economies directly to the industry.

Cobb-Douglas production function

The original production function developed by Cobb-Douglas was $Q = A L^\alpha K^\beta m$

Where Q is quantity of output, A is constant, L is the labour input, K is the Capital input, m is the error item or disturbance factor. α and β are the coefficients of labour and capital respectively.

α and β also represent the factor shares in output. Cobb-Douglas assumed that $\alpha + \beta = 1$ i.e. constant returns. Though they assumed constant returns it can as well be used for increasing or decreasing returns. If $\alpha + \beta > 1$ increasing returns operate and if $\alpha + \beta < 1$ diminishing returns operate.

If we express the production function as $Q = A L^\alpha K^\beta$.

Marginal productivity of labour and capital can be derived through partial differentiation

$$\text{Where } MP_L = \frac{dQ}{dL} = \alpha \cdot A L^{\alpha-1} K^\beta = \alpha \cdot (A L^\alpha K^\beta)^{L-1}$$

$$\alpha = \frac{Q}{L} = \alpha \cdot (AP_L)$$

$$\alpha = MP_L / AP_L$$

Where MP_L is marginal productivity of labour and AP_L is average productivity of labour

Similarly we can as well obtain the Marginal productivity of capital i.e. MP_K

$$MP_K = \beta \cdot Q / K$$

$$\beta = MP_K / AP_K$$

Cobb-Douglas production function was developed on the following premises

1. There are only two factors of production viz. labour and capital
2. The law of constant returns operates
3. The production function is a multiplicative production function hence the value of all the variables must be positive
4. Cobb-Douglas production function though is non linear it can be converted into a linear function using logs.

LAWS OF RETURNS

The Law of returns is also known as the law of variable proportions or law of diminishing marginal returns.

The law of variable proportions is an integrated law comprising three other laws namely

- a) The law of increasing marginal returns
- b) The law of Constant marginal returns
- c) The law of diminishing marginal returns

The law of variable proportions state that out of the four factors of production if some factors are held constant and vary the other factors of production the marginal productivity from

every additional dose of the variable factors increases for some time then remains constant and finally it diminishes. The law does not specify how many factors are to be kept constant and how many are variable. It simply states that there should be some fixed factors while others are variable, which means one factor can be kept constant or three factors can be kept constant. In other words, there should be some fixed and some variable factors of production.

Assumptions

The law of variable proportions is based on the following assumptions

- 1) **Variability of factor proportions:** Generally the factors of production are to be employed in a fixed proportion and there is no chance of deploying them in whatever manner one likes. The economic theorists on the other hand assumed that the factor proportions could be varied in whatever manner they like and that there is no rigid relationship between them.
- 2) **Homogenous variable factor units:** The various units of the variable factors are assumed to be homogenous i.e. identically the same. In other words there is no difference in the qualities, efficiency and ability of one variable factor unit over the other. For example if labour is the variable factor, all laborers are assumed to be identical in their physical and mental attributes and abilities. This assumption is essential to explain the fact that the variations in output are the result of some other reason other than the changes in the qualities of variable factor units.
- 3) **Absence of technical progress:** It is assumed that there is no technological progress. The techniques of production remain the same through out the production process and production period.

of factors of production...
variable proportions rushes out of the window".

LAW OF RETURNS SCALE OR LONG RUN PRODUCTION FUNCTION

The law of returns to scale states that if all the four factors of production are increased in a definite proportion the resultant marginal productivity increases at a more than proportionate rate for some time, at an equi-proportionate later on and ultimately at a less than proportionate rate.

Like the law of variable proportions the law of returns to scale is also an integrated law consisting of three important laws called as the law of increasing, constant, and diminishing marginal returns to scale.

The long run permits the adjustments in all the factor inputs. There is no difference between fixed and variable factors as in the case of the law of variable proportions. In other words changing all the factor proportions at a definite rate alters the size or the scales of plant and machinery. The resultant output however increases at an increasing, constant and diminishing rate as illustrated in the following table 5.3 and figure-5.12.

Table-5.3

Doses of factors of production	Marginal productivity	
1	10	Law of increasing Returns to scale
2	15	
3	20	
4	20	Law of constant returns to scale
5	20	
6	15	
7	10	Law of diminishing returns to scale
8	5	

It should be observed that as additional doses of the inputs are employed the marginal productivity increased at a more than proportionate rate till the 3rd dose indicating the operation of law of increasing returns to scale. From the 3rd dose to the 5th dose the marginal returns remained constant there by giving rise to the law of constant marginal returns to scale. From the 6th dose on words marginal productivity started diminishing indicating the operation of the law of diminishing marginal returns to scale. This can be illustrated through the figure-5.12

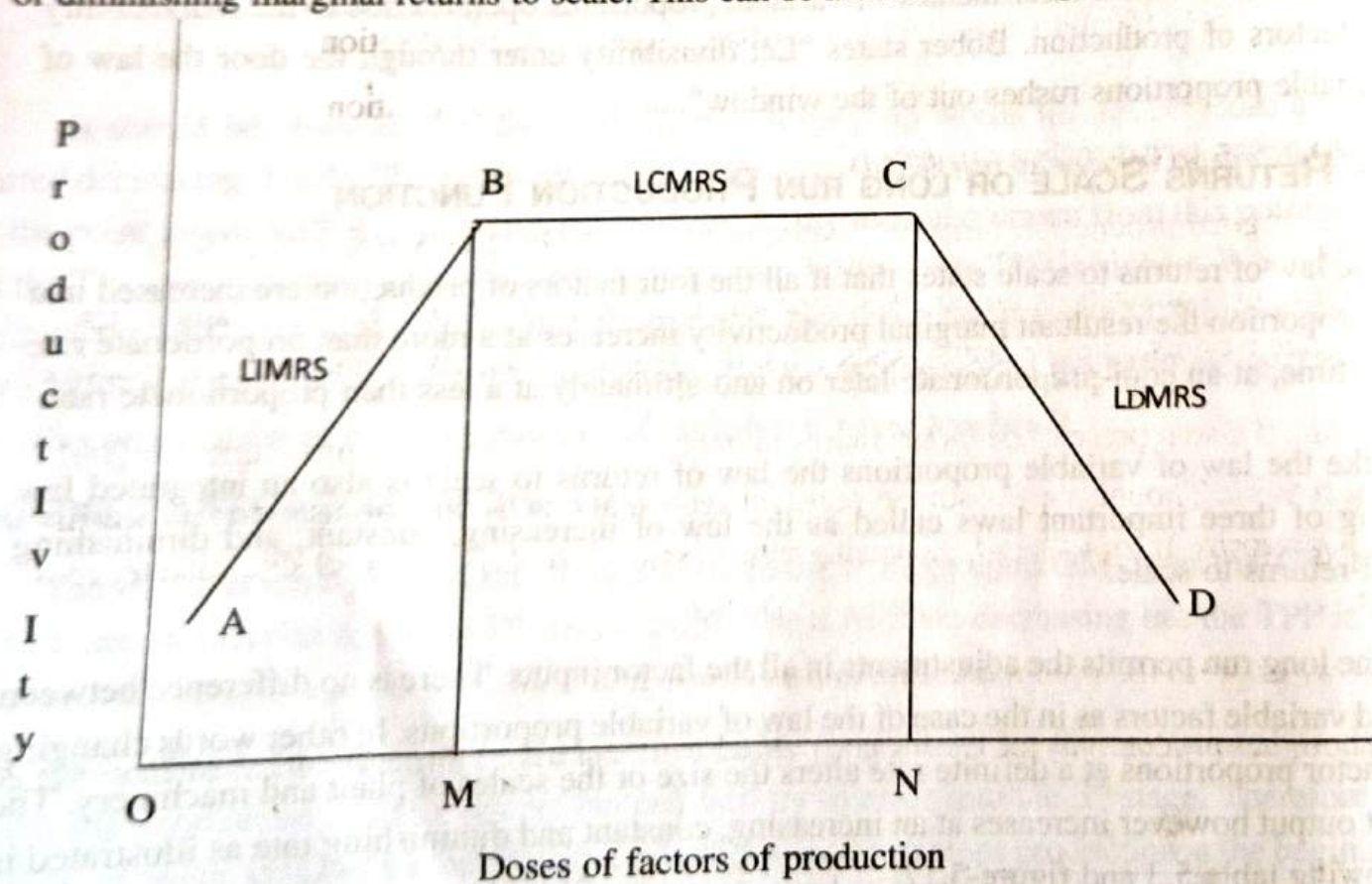


Figure-5.12

INTRODUCTION

Costs are the expenses incurred for producing goods and services. Understanding costs and analysing the cost behaviour is one of the most complex exercise for a business decision maker.

Costs are important as they have wide ranging implications on pricing policies, revenue yields, profits, wage, dividend policies and so on.

CLASSIFICATION OF COSTS

1. Nominal and Real costs.

Nominal Costs : Nominal costs are also known as money costs. Generally the business firms estimate their costs in terms of the amount of money expended on raw-materials, wages paid to labour, interest on capital invested, material costs etc. Nominal costs are the universally accepted measure of estimating and analysing the cost behaviour.

Real Costs : Real costs on the other hand are the actual amount of services of the factors of production expended in the production of commodities. For example they are measured in terms of the number of man-hours of labour time used up and the number of hours of machine time required for manufacturing goods and services. The Real costs remain absolutely stable forever. To manufacture a table the amount of wood required, the labour-time and machine -time required remains unchanged at any point of time. The comparative cost advantage theory propounded by David Ricardo is an example of a theory based on the real costs.

Unfortunately Real costs cannot be aggregated or added up. Therefore Economists prefer to analyse the cost behaviour based on the nominal costs or money costs of production. In economics here after where ever the concept of cost is used it implies money costs only.

making process.

4. **Direct and Indirect costs or Traceable and Common Costs :** Direct costs are costs that can be directly attributable to a particular product. Such as labour cost, material cost, transport cost etc. But there are certain costs, all of which cannot be directly attributable to any

single product or process. Those costs are called indirect costs. For Example HMT produces different commodities like machine tools, watches, bulbs etc. but it has a common administrative departments and personnel. The expenditure incurred on those departments or wages or salaries paid cannot be traced to one product, so like selling costs. These costs are called indirect costs. However the cost accountants try to distribute them amongst the different products in proportion of the value of output of each product produced.

5. **Past and Future costs :** Costs that are already incurred over the years and recorded in the account books already are called past costs. These costs are maintained only for the purpose of record keeping may for reviewing the previous records and in order to take or plan what costs are to be incurred or what are to be avoided.

Future costs on the other hand are costs that are likely to be incurred. The projection of the future costs has lot of managerial significance as the business unit can plan its resource mobilisation and financial planning, cost control, pricing decisions, add or drop decisions etc.

6. **Accounting costs and Economic costs :** Accounting costs are costs that are recorded in the account books of the company. The accounting costs include direct material cost, labour cost, power; water, transport charges etc. Economic costs include both accounting costs and implicit costs of self-owned factors of production.

Out of pocket costs and book costs

Out of pocket costs involve payment of money to the agents of production. These are normally recorded in the account books. But modern corporate houses include depreciation allowance as a cost in the record books but no money is paid out. These costs are called book costs.

payouts to permanent labourers etc.

COST BEHAVIOUR

Costs play a crucial role in business decision-making and profit planning. Therefore understanding costs and cost behaviour is crucial for the business decision makers.

For the purpose of analysing cost behaviour time is divided into two periods namely Short run and Long run. It should be remembered that there is no calendar period to be called short run or long run. Short run is defined as a period in which some costs of production are fixed while others are variable. On the other hand long run is a period in which all costs of production are variable. In this sense short run could be a few days or months or years depending up on whether all costs are variable or not. Therefore what is short run or long run depends on the nature of the commodity. For consumer goods like coolers or heaters a period less than a year could be short run while for commodities like Iron & Steel whose gestation period is long the short run may extend up to 15 to 20 years. The only test to differentiate between short run and long run is whether all costs are variable or not.

NATURE AND BEHAVIOUR OF SHORT RUN COSTS

Short run is a period in which some costs of production are fixed while others are variable.

FIXED COSTS

Fixed costs are the Wages payable to the permanent Labourers, Rents on Buildings, and Interest payable on borrowed capital, minimum power and water charges, insurance premium and the minimum necessary profits to induce a producer to continue the production. The Fixed costs remain absolutely fixed irrespective of the volume of output. In the short run fixed costs cannot be increased or decreased. Even if the output is zero fixed costs will be positive. Fixed costs are also known as historical costs or sunk costs. They are the costs already incurred and cannot be changed hence become part of history hence they are called historical costs. They are irretrievable costs

because once incurred there is no chance of recovering or reducing them. They are also called as indirect costs, supplementary costs, and overhead costs.

VARIABLE COSTS

The variable costs include Wages payable to the Temporary Labourers, Raw Material, charges, Transportation costs, and power & water charges over and above the minimum.

Variable costs change with output. If the output is zero variable costs are zero and as the output increases Variable costs also increase. Variable costs are also known as direct costs, prime costs, special costs etc.

The following table- 6.1 and figure -6.1 illustrates the behaviour of Short Run costs.

Table 6.1

1 Output	2 TFC	3 TVC	4=2+3 TC	5=2÷1 AFC	6=3÷1 AVC	7=5+6 AC	8 MC
0	30	0	30	“	-	“	-
1	30	10	40	30	10.0	40	10
2	30	18	48	15	9.0	24	8
3	30	24	54	10	8.0	18	6
4	30	30	60	7.5	7.5	15	6
5	30	38	68	6.0	7.6	13.6	8
6	30	48	78	5.0	8.0	13	10
7	30	60	90	4.3	8.6	12.9	12
8	30	75	105	3.8	9.4	13.2	15
9	30	95	125	3.3	10.5	13.8	20
10	30	120	150	3.0	12.0	15	25

A careful observation of table-6.1 indicates the behaviour of short run costs of production.

The total fixed cost (TFC). The total fixed cost (col.2) is positive even when the output is zero and remained fixed irrespective of the volume of output. When this data is plotted on a graph the TFC curve runs parallel to the horizontal axis. This can be observed from figure-6.1 given below.

Total Variable COST (TVC). The total variable cost (Col.3) is zero when the output level is zero. As the output increases the TVC is increasing at a decreasing rate initially and at an increasing rate afterwards. This is due to the advantages arising out of division of labour and

BREAK EVEN ANALYSIS

INTRODUCTION

A business firm is interested in maximisation of profits. Profits are measured as the difference between Total Revenue and total cost. $\Pi = TR - TC$

Where Π = Profits, TR = Total Revenue, TC = Total Cost

Before a firm plans for profit maximisation it has to determine the Break-even Quantity or value of output it should produce. It is the minimum output that a business unit should produce in order to avoid any loss. Break-even point is defined as a point where a firm neither gets profits nor losses. It is a point of no profit and no loss. In other words break-even point is a point where firm leaves the loss zone and enters the profit zone. Breakeven analysis has wide ranging implications to a business manager. It is useful in understanding the costs, revenue, profit, capital investment decisions, production planning, add or drop decisions and so on. In the accompanying study a thorough analysis is presented regarding the break-even analysis.

BREAK EVEN ANALYSIS

Before we precede any further it is necessary to recapitulate the costs and revenue analysis that was presented in the previous lesson.

- a) Costs of production are segmented as Fixed and variable costs i.e. $TC = TFC + TVC$
- b) The Total Revenue is obtained by Multiplying Sales quantity with Price. $TR = Q \times P$ where Q is sales quantity and P is Price
- c) Accountants Perception of Costs and Revenue are different from economists Perception
- d) Break-even analysis assumes that Price remains constant and any amount of out put can be soled at the given price. Therefore the TR curve is a straight line sloping upwards from left to right observe table 7.1 and Figure-7.1

Though the accountants' perception of costs and revenue are untenable yet breakeven analysis developed by them have a lot of advantages despite their limitations in decision-making process. Break-even point can be calculated graphically as well as algebraically. Observe Figure 7.3.

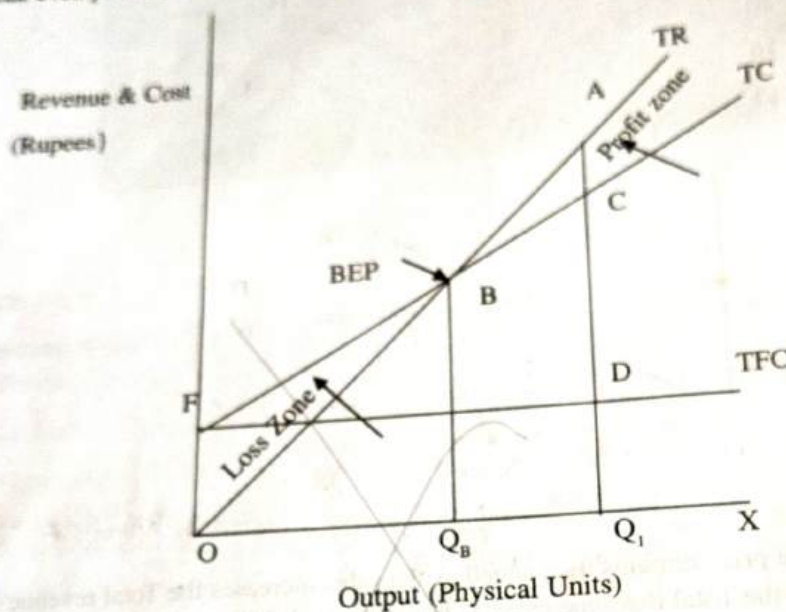


Figure 7.3

Observe figure-7.3. At output level of OQ_B the total Revenue is equal to total cost i.e. B where the firm is neither getting profit nor loss. Hence OQ_B is the breakeven quantity of output. At output level less than OQ_B the firm is suffering losses and beyond that it is reaping profits. At point B the firm is leaving the loss zone and entering the profit Zone.

In figure -7.3 at OQ_1 Output, the total Revenue is AQ_1 and total cost is CQ_1 . The net profit is AC.

The breakeven chart can be used to know what will be the breakeven quantity if the fixed cost change or variable costs change. Figures 7.4 illustrates the BEP when fixed costs change. Observe figure-7.4 when the fixed cost is TFC with TR and TC being the revenue and cost function. The Breakeven quantity is OQ. When the fixed cost raise to TFC_1 the TC curve shifted upwards and parallel to TC but the BEQ increased from OQ to OQ_1 .

MARKET STRUCTURES

Modern business units function in a competitive environment. An understanding of the nature and type of market in which a firm operates is imminent for it to determine its pricing policies, output decisions, determining the outlay for selling costs etc. In this lesson the emphasis is on classification market and the criterion to identify the markets. Remember that in the process of constructing the market models the economists made use of several assumptions regarding number and size of Sellers and Buyers, product substitutability and interdependence and the various entry barriers.

WHAT IS A MARKET?

Market is a place where commodities are bought and sold. In a market there will be two types of transactions namely sale and purchase. The market is located in a particular locality or building. For example the giant super market is located in a specific building where as the grain market in Hyderabad is located at Begumbazar, the pearl market near charminar etc.

In economics the term market is used in its wider sense, it includes a group of people or firms in contact with one another for the purpose of buying and selling.

Stonier and Hague define market as "any organization whereby buyers and sellers of a good are kept in close touch with each other. There is no need for a market to be in single building. The only essential feature of a market is that all buyers and sellers should be in constant touch with each other, either because they are in the same building or because they are able to talk to each other by telephone at a moments notice."

Samuelson defines "A market is a mechanism by which buyers and sellers interact to determine the price and quantity of a good or service"

There is no need for any physical contact between buyers and sellers and there is no need for a specific location in any one area or building.

In other words in a market there should be a) Sellers b) Buyers, C) goods or services and d) Exchange transactions.

CLASSIFICATION OF MARKETS

Markets are classified on the basis of the following features.

By Location: Local market if the goods produced is offered for sale in the same locality or region. National market in case the goods are sold within the geographical boundaries and International market where goods are exported for sale from one country by crossing geographical boundaries.

By size of business: Based on the size of business the markets are classified as wholesale and retail markets.

By time: Based on time markets are classified as very short run, short run and long run markets.

By controls: If the government regulates the prices or production it is called regulated market and if it does not interfere in anyway it is called unregulated market.

By availability: Based on the availability of the product markets are classified as Sellers market and Buyers market. If there is grave shortage of product in the market and the producer can sell the product on his own terms and conditions it is called Sellers market. If the products are tuned to suit the needs of the consumers it is called the Buyers market.

MARKET STRUCTURES

According to Pappas and Hirschey "Market structure refers to the number and size distribution of buyers and sellers in the market for a good or service"

Markets may be classified based on the following criterion

- A. The number of sellers and buyers
 - B. The nature and substitutability of the product
 - C. Existence of barriers to entry
- A. Based on the Number of Sellers of Buyer

1. Based on number of sellers

Based on the number of sellers the markets are classified as perfect market, Monopoly, Duopoly, Oligopoly, and monopolistic markets.

- i) **Perfect market:** Under perfect market there are a large number of firms producing fairly insignificant quantity of the product in the market. Since the number is large

their decisions regarding the quantum of output to be produced has no bearing on the market supply and also the price. An individual firm is a price taker but not a price maker. It is an extreme form of market.

- ii) **Monopoly:** In this market a single producer controls the entire supply of the product. There is no difference between firm and industry. The entire supply of the commodity is under the control of a single producer. He is the price dictator. It is another extreme form of market.
- iii) **Monopolistic market:** In this market there are a number of firms producing heterogeneous products. An individual firm has little or no influence on the market supply but due to product heterogeneity it can independently determine the price.
- iv) **Oligopoly:** It is a market where there are few firms may be 10 or 9 producing either homogenous or heterogeneous products. Each firm in this market will have a sizable control over the market supply. But due to collusion or price leadership prices a firm may not independently determine the price.
- v) **Duopoly:** Duopoly is a market where there are only two sellers each one having significant control over the supply of the market and also the price.

2. Based on number of buyers

Markets are also classified based on the number of buyers

- i) **Monopsony:** It is a market where there is a single buyer of the entire product. Since there is a single buyer he can get the products at considerably low prices and the suppliers tailor the product to the needs of the buyer.
- ii) **Duopsony:** It is a market where there are only two buyers of the product. Each will have considerable influence on the price.
- iii) **Oligopsony:** It is a market where there are few buyers of the product and with some degree of control over the price.

B. Product substitutability & interdependence

The market structures can also be classified based on the basis of product substitutability and product interdependence.

1. Substitutability

The degree of control that a firm can exercise on price is influenced more by the substitutability of the products than the number of producing units. The degree of substitutability can be measured by the cross elasticity of demand. The cross elasticity can be measured by the following formula.

2. Interdependence

Product interdependence can be used to measure the reaction of the competitors. The degree of product interdependence can be measured by the cross elasticity of quantity. This measure is useful in understanding the impact on the price of one firm due to an increase in the output by the other firm. The degree of interdependence can be measured by the following formula

C. Entry barriers

The nature of the market structure is influenced by the ease of entry and exit.

In fact a large number of entry barriers exist preventing the entry of new competitors into the market. Some of the major barriers to entry are.

- i) **Legal restrictions:** Taking into consideration the strategic needs and security threats the government prohibits the entry of new firms in certain fields of activity through statutory controls. This acts as a strong barrier on the entry of firms into the industry. Most of the monopoly public sector enterprises come under this category.
- ii) **High initial investment:** Some industries require large doses of investment and are capital lumpy in nature. The quantum of investment may run into hundreds or thousands of crores of rupees. Few firms may be able to mobilize such large doses of funds hence it acts as a strong entry barrier.
- iii) **Gestation period:** Firms desire quick returns on investments. If the gestation period of conceiving and completing a project is long firms may not try to enter into those product lines for example Iron and steel industry.
- iv) **Patent rights:** Some Products are controlled by patent rights as a result entry into those product lines become very difficult. Entry becomes more difficult if the product involves complex technology.
- v) **Switching costs:** The users of larger machines and complex machines do not want to change their brands for the fear that they have to incur more costs on training the labour and maintenance. On the other hand they prefer to use known and familiar brands as it saves the switching costs. Thus the switching costs also act as an entry barrier.
- vi) **Price and profits:** The entry of new firms is also governed by prices and profits. Higher profits induce new firms to enter into the industry and lower profit margins prevent the entry.

PERFECT COMPETITION

The economists assumed that perfect competition exists in markets. In such a market there are large number of buyers and sellers of an identical or homogeneous product having no transport cost and the price of the product is determined by the free working of market forces i.e. demand and supply. There is single price that prevails in the market all firms sell the product at prevailing price. In this market the seller is a price taker. According to Mrs. Joan Robinson "perfect competition prevails where the demand for the out put of such product is perfectly elastic. This entails first that the number of sellers is large so that the output of any one seller is a negligiely small proportion of the toal output of the commodity and second that the buyers are all alike in respect of their choice between rival sellers so that the market is perfect." It exists when the following conditions are satisfied.

FEATURES OF PERFECT COMPETITION

1. Large number of buyers and sellers
 2. Homogenous product
 3. Free entry and free exit
 4. Perfect knowledge about the market
 5. Absence of transport cost
 6. Absence of advertisement
 7. Perfect mobility of factors of production
1. **Large number of buyers and sellers :** Under Perfect Competition the number of sellers and buyers is large, a single firm produces an insignificant portion of the total supply in the market, it cannot influence the market supply on its own. Similarly a single buyer purchases so little he cannot influence the market demand for the commodity. Hence the price is determined at the intersection point of the Market demand and supply curves. A firm is only a price taker but not a price maker. At the Market given price a firm can sell any number of units. Therefore the Average Revenue (AR) and Marginal Revenue (MR) curves coincide with one another and are parallel to the 'X' axis as shown in figure 8.1

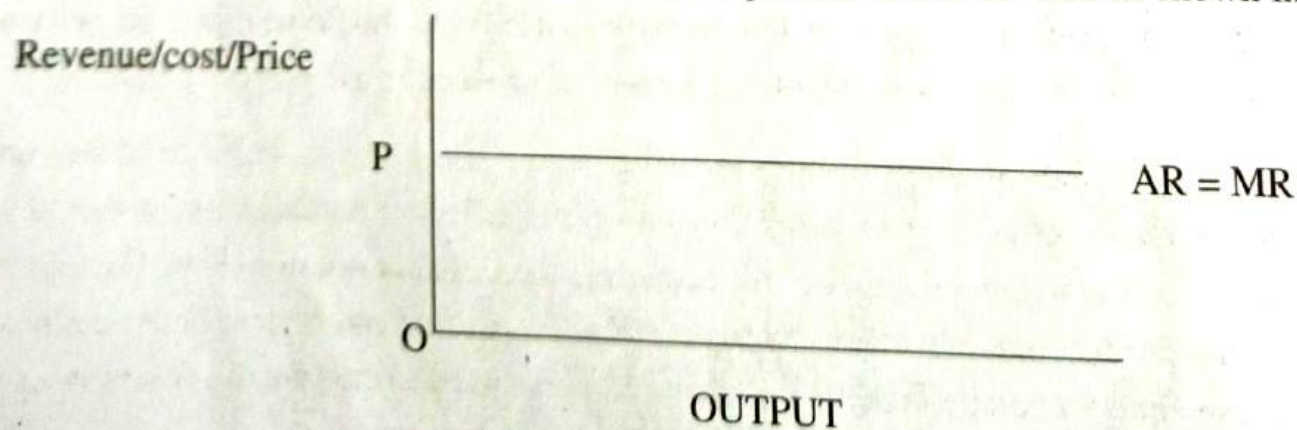


Figure-8.1

2. **Homogenous product:** Under perfect competition all the commodities produced by all the firms are identically the same or standardized. There are no differences, real or imaginary, among the products. The products manufactured by all the firms are perfect substitutes for one another. Hence no seller can charge a higher price or sell at a lesser price. The same price rules through out the market.
3. **Free entry and exit:** There are no barriers to the entry and exit of the firms. Firms enter or leave the industry based on the profits or losses they get. When the existing firms are getting abnormal profits in the short run new firms are induced to enter into the industry in the long run. With the entry of new firms the supply increases and the price falls. As a result the abnormal profits disappear in the long run. On the other hand if the firms in the short run are suffering losses, some firms leave the industry in the long run thereby reducing the market supply. The price therefore tends to rise and the losses disappear. Thus under perfect competition firms may get abnormal profits or losses in the short run but in the long run they should get only normal profits.

If a market satisfies the above **three** features it is called **pure competition**. If competition is to be called perfect then the market should satisfy the following features also.

4. **Perfect Knowledge:** Under perfect competition it is assumed that both the buyers and sellers have perfect knowledge about the market. Hence the same price rules throughout the market.
5. **Absence of Advertisement:** Since products are homogenous and the consumers have perfect knowledge about the product there is no need for advertisement and no purpose will be served by advertisement.
6. **Absence of Transport costs:** The golden rule of perfect competition is that the same price has to prevail in the entire market. If the price is to remain the same then there should not be any cost of transportation. In other words a large number of small firms come up at different locations catering to the needs of the buyers and does not need any transport costs.
7. **Perfect mobility of factors of production:** Under perfect competition all the firms should charge the same price. If the price is to remain the same costs should be equal. In order to eliminate cost variations it is assumed that factors of production are perfectly mobile.

INTRODUCTION

The word Monopoly is derived from two words mono and poly. Mono means single and poly means seller. Monopoly is a market where there is a single seller dominating the entire market without there being any close substitute. Monopoly is one of the forms of imperfect market, the others being duopoly, oligopoly and monopolistic competition.

FEATURES OF MONOPOLY

1. Single seller and large number of buyers
2. Homogenous product
3. Absence of close substitute
4. Restrictions to the entry of new firms

A market that satisfies the above features is called monopoly.

- 1) **Single seller:** In monopoly there is a single seller. There is no difference between the firm and industry. The entire market supply of the product is under the control of the monopolist. He can increase or decrease the supply of the commodity in the market at his own discretion. The monopolist is a price maker or a price dictator. Thus the monopoly firm has control over the price as well as supply, but it can determine only one of them and leave the other to the market. If the monopolist determines the supply then what price is to be paid depends on the market demand. On the other hand if he determines the price what quantity is to be produced should be left to the market. Since, the price and quantity demanded are inversely related the demand curve faced by a monopoly firm slope downwards from left to right. When the demand or the average revenue curves slopes down the marginal revenue curve also slopes downward but always below the average revenue curve as shown in figure-9.1.

Large number of buyers: Under monopoly the number of buyers is many. Therefore no single buyer has any control over the market demand or the price of the commodity¹.

Homogenous products: Since a single firm controls the entire market all the commodities produced by it are identically the same. Therefore a monopolist under all normal conditions charges the same price².

Absence of close substitutes: There is no close substitute for the monopoly product. If there is no substitute at all, then that monopoly is called pure monopoly. In case there is a substitute but not a close substitute then it is called simple monopoly. In the absence of a close substitute the cross elasticity of demand is less. Hence the monopoly faces a demand curve, which is relatively inelastic or <1 . Therefore average revenue curve (demand curve) steeply slopes downwards.

Restrictions to the entry of new firms: under monopoly there are restrictions to the entry of new firms in to the industry. These restrictions may take the form of a) legal b) Artificial and C) Natural restrictions

- a) **Legal restrictions:** These are the restrictions imposed legally by the statutory authority prohibiting competition in certain lines of activity. These restrictions are generally imposed based on strategic and security considerations. For instance the industrial policy resolutions announced by the government of India reserved the manufacture of certain products like arms and ammunition, atomic energy etc. for the public sector and private competition is prohibited. These are normally called as Public monopolies. Legal monopolies also come in to existence in the private sector through the grant of patent rights and copy ship rights to the inventors and authors respectively. Legal monopolies are the most justified forms of monopoly.
- b) **Artificial restrictions:** These are the most dangerous forms of monopolies. They emerge due to the following reasons.

i) **Cut throat competition.** When a new competitor enters the market the monopoly slashes down the price to such low levels that no new competitor can survive in the market.

ii) **Exclusive control over raw material:** If a single firm has exclusive control of a critical raw material to produce a commodity, it acquires monopoly power and does not like to share it with a new competitor. Thus the firm gets monopoly power.

iii) **Dumping:** Dumping is a method normally resorted to by the multinational companies. Flooding the foreign markets with home produced products and selling them in the foreign markets at prices much lesser than the home prices is popularly known as dumping. The objective behind dumping is to acquire monopoly power over the supply of a commodity in the markets abroad.

iv) **Unfair practices:** Some unscrupulous producers resort to illegal and unfair methods to acquire monopoly power. Creating labour troubles in the competitors firm or cornering the raw material, stoppage of credit flows, stay out and boycott arrangements etc. are few examples of unfair practices. However these practices are declared as illegal by most of the countries.

C) **Natural Restrictions:** These are the monopolies that arise due to the limited size of the market. A single firm operating at the optimum level can meet the entire market demand hence there is no scope for another competitor. These are also called as economic monopolies. Due to the above restrictions a single firm acquires monopoly power. 3)

... AS A RISK OF governmental interference under monopoly.)

MONOPOLISTIC COMPETITION

It is market characterized by an admixture of the features of both monopoly and perfect competition.

Features of Monopolistic Competition

I) Number of buyers and sellers

II) Heterogeneous products

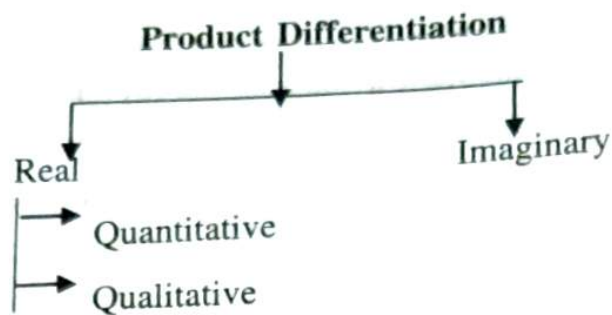
III) Free entry and Exit

IV) Close substitutes

V) Selling costs

i). **Number of buyers and seller:** There are large number of buyers and sellers in the market. Hence no individual firm can influence the supply. Often the presence or absence of a firm goes unnoticed by the market. So is the case with buyers.

ii). **Heterogeneous products:** Unlike perfect competition each producer under monopolistic market produces differentiated products.



The differences in the products can be classified as **real and imaginary** differences. Real differences may be either **quantitative or qualitative**. Quantitative differences may take the form of variations in weight, length, size etc. Qualitative differences reflect changes in the quality of product or service in some respect or the other. Imaginary differences are created by the producers in the mind of the consumers through advertisement or salesmanship.

It is due to the product differentiation Monopolistic firms acquired the capacity to determine price independently. Hence a firm under monopolistic market is a price maker but not a price taker. But it should be borne in mind that the demand curve (average revenue curve) faced by monopolistic firms slope downwards from left to right, indicating the inverse relationship between price and quantity demanded.

- iii) **Free entry and Exit:** Like perfect competition under monopolistic market also firms have the freedom to enter or exit the industry. When existing firms are making abnormal profits new competitors are induced to enter the industry and in case the existing firms are suffering losses some of them may leave the industry. This implies that firms under monopolistic markets can get only normal profits in the long run.
- iv). **Close Substitutes:** Under monopolistic markets firms produce goods that are close substitutes to each other's products. Hence the cross elasticity of demand is greater than unity. Therefore monopolistic firms face a more elastic demand curve.
- v). **Selling Costs:** Monopolistic firms incur huge amount of selling costs. Selling costs are costs intended at promoting the sales. These costs include advertisement expenditure, free gifts, dealer discounts, hospitality expenses etc. The most important component of selling costs of course is advertisement expenses. The objectives of advertisement are a) Creating consumer awareness b) attracting new customers and c) retaining the existing customers. A good advertisement should be unique, appealing and sensational. While incurring advertisement firms must be cautious and should bear in mind that there is no proportionate relationship between advertisement and sales, benefits of advertisement may go to rivals and where consumers are brand attached advertisement may not be effective.

PRICING STRATEGIES

INTRODUCTION

Pricing is a crucial but complex component of firm's decision-making process. In the determination of prices a firm has to take a number of factors and variable into consideration including of course the costs of production. The demand supply forces may not always fit in to the determination of prices as the firms operate in imperfect markets, with insufficient information both about the market as well as government policies. The following analysis through some light upon the various objectives, and methods of pricing

PRICING OBJECTIVES

Though economic theory assumes that every firm tries to maximise its profit. In reality firms may pursue altogether different objective like sales maximisation, or growth maximisation and so on. Therefore the pricing objectives of a firm depend upon its overall business objective. Understanding the broad pricing objectives is essential before making an in-depth analysis of pricing policies and strategies. The pricing objectives are listed below

1. **Profit maximisation :** The objective of any business enterprise is to make profits. Traditional economic theory believes firms try to maximise profits. This may be true in case of owner managed business enterprises. In the realistic market situations characterized by a number of complexities, operating under conditions of inadequate information and uncertainties firms may not be in a position to maximise profits.
2. **Sales maximisation :** Most of the modern business enterprises are corporate bodies where management is divergent from ownership. The managers of these business units may try to maximise sales and obtain market domination by penetrating their sales. In such cases firms determine prices at such levels where they try to maximise sales rather than profits
3. **To attain a fixed rate of return on investment :** Some business firms try to determine the prices at such levels where they try to get a certain amount of fixed returns on their investments at least in the long run perspectives.
4. **Maintain product distinctiveness and brand image :** Firms may try to fix prices in such a way where they try to maintain their product distinctiveness. Some of them may fix high prices to prove the high quality of their product and try to somehow establish that their products are different from those of the other products in the markets. Prices are fixed by some of the firms to build up their band image. For example branded products have higher price tags than unbranded products
5. **To prevent governmental interference :** Some producers fix reasonable prices for their products in order to avoid the wrath of government declaring them as exploiters and to avoid unwanted interference by governmental agencies

6. **To prevent the entry of potential competitors :** Normally high prices and profits in one business line invite entry of new competitors in the long run. Therefore to prevent the entry of competitors even the monopoly firms try to charge low prices.
7. **To avoid price war :** This is an extreme case of oligopoly units where rival consciousness is high and any reduction in the price of a commodity leads to competitive price cuts and thus creating price war. Price war is always destructive to all the firms. To avoid the price war oligopoly firms try to follow the prices fixed either through collusion or price leaders.
8. **To accept Social responsibility :** Modern business enterprises are now considering that they have a social responsibility. They feel their business is not just to make profits but also serve the society in some form or the other. Such firms may charge a low price than a profit maximising one.
9. **To maximise Long run profits than short run profits:** Firms that are interested in long run profits than short run profits may charge low initial prices that will prevent competition and entry of new firms into that product line. This brings in an added advantage to maximise their long run profits.

PRICING METHODS

There are a number of pricing methods or practices that are adopted by business enterprises. They can be grouped in to three categories viz.

- A. Cost oriented pricing
- B. Competition oriented pricing
- C. Other Methods

A. COST ORIENTED PRICING METHODS

Cost oriented pricing methods are classified as the following three methods

1. Cost Plus Pricing:

Cost plus pricing is also known as Full cost pricing or Standard pricing.

R.L Hall and C.Z.Hitch developed the concept of cost plus pricing in the year 1939.

Cost plus pricing assumes that the costs are fixed and variable costs are easily ascertainable and sales volume is known. The top-level managers determined a certain markup on the costs of production based on the business objectives of the firm. This markup is the net profit margin for the firm over the costs

According to this method the price is determined as follows

Assuming

2. Incremental cost pricing or Marginal cost pricing

Marginal cost is an addition made to the total cost for unit change in production. In reality firms do not change their output by unit. Normally there are chunk changes in the level of output produced. Here in comes the Incremental principle. Incremental costs or revenues are calculated based on the chunk changes in output. From the incremental costs and revenues one can calculate the marginal cost and marginal revenues.

B. COMPETITION ORIENTED PRICING

1. **Going rate pricing:** Some firms adopt a price ruling in the market rather than independently determining the price on their own. This is called Going rate pricing. The advantage of this pricing is it saves the firm from the ordeal of collecting cost data also avoid the risk of uncertainty it may face in the event it determines its own price. Oligopoly firms normally adopt this method when the price leadership is dominant. Under these conditions they resort to product tailoring i.e. adjusting the costs with the prices.
2. **Loss leader pricing :** This is a method adopted by firms producing joint products or supermarket. Loss leader is a prime product that contributes less to the net profit margin compared to the other product or products in its product line. For example super markets fix relatively low prices on food grains and pulses and recover what ever the low margins they get by pricing other products at a higher level. Remember that a loss leader need not bring losses but it may contribute less relative to other products. The points that are to be kept in mind here are a) That the price of the loss leader should be substantially lesser and the buyer should buy substantial quantity to feel the advantage of buying the product and it there should be no compromise on the quality of the product.
3. **Trade association pricing :** Trade associations generally determine prices in order to avoid a price war among the members. All the manufactures generally charge the same price. These prices are generally determined through explicit or tacit collusion. Remember that explicit collusions are declared illegal under the anti-trust laws all over the world and competition act in India.
4. **Customary pricing :** It is a method of pricing goods continuously at a particular level and affecting any change in their prices even if the costs escalate. In this method the firms try to adjust size, quantity, or quality of the product but never change the price
5. **Price leadership :** It is a method of oligopoly pricing where there exists price leadership. The price leader determines the price and others simply follow the price fixed by the price leader. The price leadership is classified as
 - i. **Dominant Price leadership:** It is a dominant firm that controls a lions share of the market sales that determines the price while others follow the price leader.
 - ii. **Barometric Price Leadership:** An old or well-established firm determines the prices while others follow. This is also called as democratic price leadership.

ACCOUNTING

Accounting is very often referred to as the business language. If you are aware of accounting, it becomes easy to understand how a business is operated.

In another sense, accounting can also be considered as a system of providing financial information, with the help of financial statements, to the interested parties.

Definition :- According to the American institute of certified public accountants, accounting is

“an art of recording, classifying and summarising in a significant manner and in terms of money, transactions and events which are, in part atleast, of financial character and interpreting the results thereof”.

Thus accounting deals with

- a) Recording financial transactions
- b) Classifying them under different heads
- c) Summarising the transactions
- d) Analysing them for a purpose.

ACCOUNTING AND BOOK KEEPING

Book keeping deals with recording of financial transactions and maintaining a record of all the transactions. In a way accounting starts, where book keeping ends.

According to Northcott, “Book keeping is an art of recording in books of account the monetary aspects of commercial or financial transactions”.

Functions of Accounting:

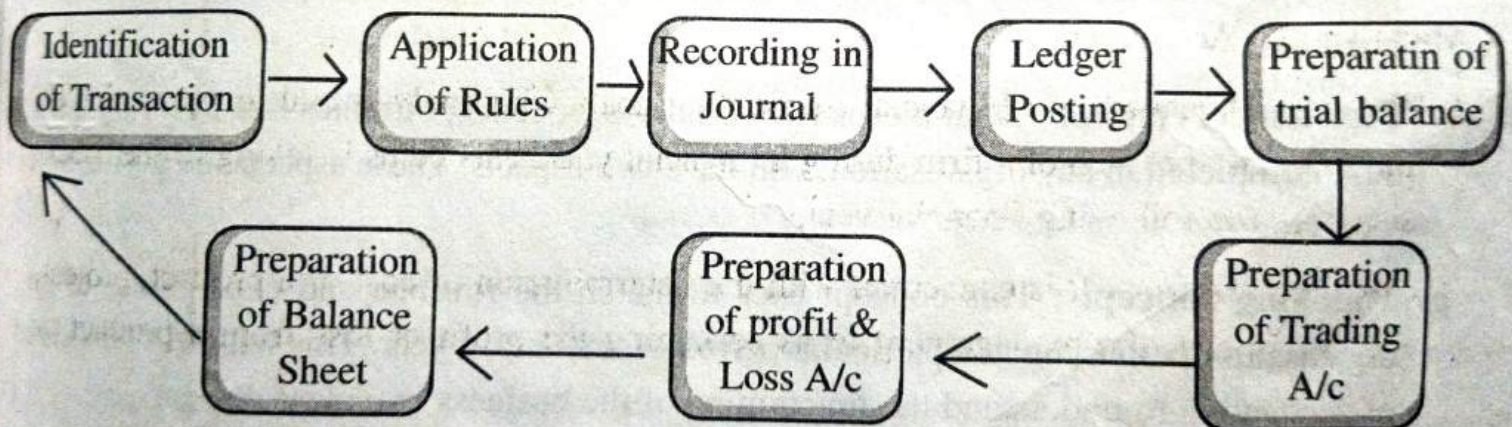
- **To Record Business Transactions:** The main functions of accounting are to keep systematic record of business transactions.
- **To Indicate the Financial Position:** Preparation of Final Accounts enables the businessman to know the earning capacity and financial position of the business unit. It shows the business capacity to others.
- **To Assist Management:** Accounting provides information to the management which helps it to take managerial decisions.
- **To Shows True Position:** It gives the true position of the business as a whole.
- **To Satisfy Government Officers and Comply Legal Requirements:** Reliable information regarding profit or loss may be obtained from the accounting records. This provides satisfaction to the taxation authorities. In case of registered companies, auditing is compulsory. Without this accounting, there is no meaning for auditing. So, accounts are to be prepared according to the requirements of the Companies Act, 1956.

Advantages of Accounting:

- **Provides for Systematic Records:** Since all the financial transaction is recorded in the books, one need not rely on memory. Any information required is readily available from these records.
- **Facilitates the Preparation of Financial Statements:** Profit and Loss account and Balance Sheet can be easily prepared with the help of the information in the records. This enables the business man to know the net result of business operations during the accounting period.
- **Provides Control Over Assets:** Book-keeping provides information regarding cash in hand, cash at bank, stock of goods, accounts receivable from various parties and the amounts invested in various other assets. As the Trader knows the values of the assets he will have control over them.
- **Helpful to Management:** Accounting is useful to the management in various ways. It enables the management to assess the achievement of its performance. This weakness of the business can be identified and corrective measures can be applied to remove them with the help of this accounting only.
- **Documentary Evidence:** Accounting records can also be used as evidence in the court to substantiate the claim of the Business. These records are based on documentary proof i.e., vouchers etc.,
- **Comparative Study:** One can compare the present performance of the business with that of its past. It gives useful conclusions and makes proper decisions.
- **Tax Matters:** Properly maintained book-keeping records will help in the settlement of all tax matters with the tax authorities.
- **Provides the Required Information:** Interested parties such as Owners, Lenders, Creditors etc., get necessary information at frequent intervals.

ACCOUNTING CYCLE OR PROCESS

This cycle of accounting begins with identification of the transaction. After identification of it is recorded in Journal according to the rules of double entry book keeping and then posted into Ledger. At the end of the financial year from these ledgers, trial balance is prepared. This trial balance helps in the preparation of trading, profit & Loss a/c and Balance sheet. This process gets repeated every year. Therefore, it is called as "accounting cycle".



Branches of Accounting:

The role of Accounting has changed from that of a mere record keeping during the 1st decade of 20th Century to the present stage, when it is accepted as information system and decision making activity. The Accounting is used depending upon the utilization of that. It is mainly divided into three branches. They are:

1. **Financial Accounting:** The basic purpose of financial accounting is to prepare the financial statements: Trading and Profit and Loss Account and Balance Sheet. The Trading and Profit and Loss account is also called as Income Statement: it shows the Net profit and Net loss of the Business Firm for the end of a given Period. The Balance Sheet reveals the financial position of the business firm in terms of its assets and liabilities as on a given date.
2. **Cost Accounting:** The purpose of this branch of accounting is to ascertain the cost of product and costs incurred for carrying out various activities. It also assists the management in controlling the costs at each level. The tools and techniques of Cost Accounting include Opportunity Cost, Marginal Costing, BEP analysis...etc.
3. **Management Accounting:** Its aim is to assist the management in taking correct policy decisions and to evaluate the impact of its decisions and actions. Management accounting is concerned with presentation of accounting information in the most useful way for the Management. The data required for this purpose are drawn generally, from Financial and Cost accountancy.

Journal:

- ★ Journal is the first book in which transactions are recorded in a chronological order (Date wise).
- ★ It is also called 'Day book'.
- ★ The following is the format of a journal

Date	Particulars	Ledger Folio	Debit (Rs)	Credit (Rs)

- ★ Recording entries in journal is called Journalizing.
- ★ The following are the benefits of writing entries in journal.
 - To show all the necessary information about a transaction.
 - To know why it is Written
 - To know the date wise transactions.
 - To help identify and prevent errors.

Ledger:

- ★ Ledger is a book that contains several accounts.
- ★ The process of preparation of accounts from the journal into ledger is called posting in the Ledger. **Example:** Sales Account, Purchase Account, Sales Returns Account, Purchase Returns Account, Debtors Account, Creditors Account..Etc.

' T ' Format of Ledger:

The Format of Ledger Account is of Two Parts:

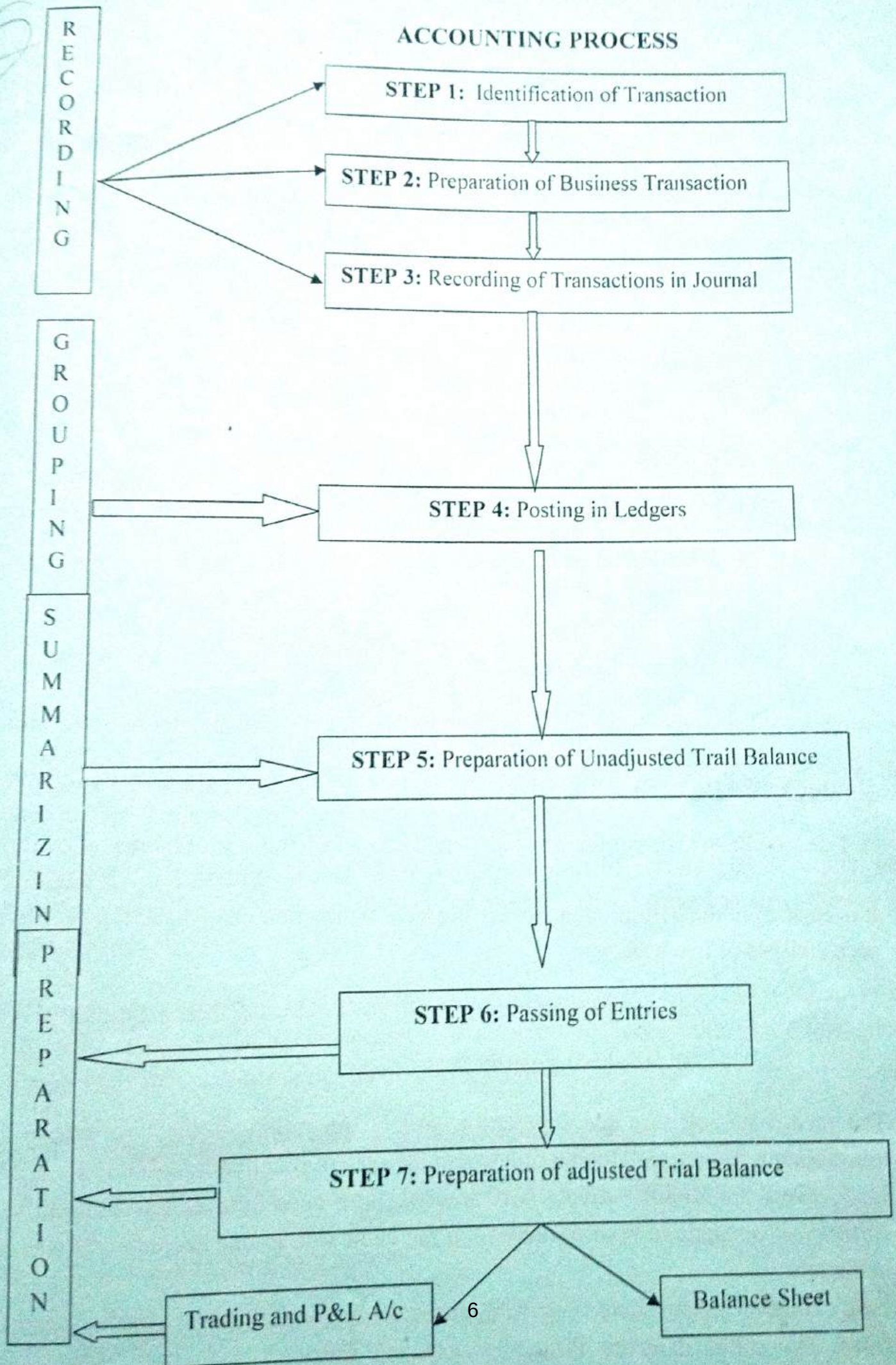
- Left Hand side called Debit side (Dr) and
 - Right Hand side called Credit side (Cr).
- ★ Debit Side start with 'To' and Credit Side start with 'By'.
 - ★ Calculate the totals on the each side of the account. Check up on which side the total is higher.
 - ★ Put the total on the both sides of the account First. Then see by how much the other side is lesser. Put the difference on the lesser side writing clearly balance carried down.
 - ★ This is the closing balance for the end of the given period.
 - ★ The same difference becomes the opening balance for the next period.

Dr				Account				Cr			
Date	Particulars	Folio	Amount	Date	Particulars	Folio	Amount	Date	Particulars	Folio	Amount

Trail Balance:

- ★ Trail Balance is a Statement containing Debit and Credit balances of various Accounts taken out from Ledger Books as on a Particular Date.
- ★ Trail Balance is prepared to ensure that there are no arithmetic errors in the Books of Accounts.
- ★ The Trail Balance must agree as on a given date i.e. The Total Debt Balances must be equal to the Total Credit Balances.
- ★ If it does not agree that means there are certain ⁵Arithmetical Errors in the Books of Accounts.

ACCOUNTING PROCESS



Book Keeping:

- ★ The main object of any Business to make profits. It may be a Business engaged in the purchase and sale of good or it may be engaged in the production of goods or provision of services. Whatever be its nature, the main object is to earn profits.
- ★ A Businessman enters into business in order to earn profits. He starts his business with a certain amount. It is called a Capital.
- ★ He buys and sells goods either for cash or on credit. In the course of his business he may have to incur some expenses for carrying-on his trade.
- ★ Every day he has to pay cash to the suppliers of goods or his employees or others who have rendered some services to him. Similarly, he receives cash from is customers and others. All these are called "**Business Transactions**".
- ★ Book-Keeping is the system of recording business activities for the purpose of providing reliable information to the owners and managers about the state and prospects of the business concern."
- ★ The system of Book-Keeping could be described as a "System of Information". Book-Keeping is a narrow term concerned mainly with the maintenance of the books of account. It covers the following four main aspects:

- i) Identifying the transactions and events to be recorded.
- ii) Measuring them in terms of money.
- iii) Recording them in the books of prime entry.
- iv) Posting them into Ledger.

- ★ Book-Keeping is restricted to level of work. Clerical work is mainly involved in it. But, by Book-Keeping it is not possible to know the final result of business every year.

Double Entry Book Keeping System:

- ★ A Transaction has been defined as that which involves a transfer of money or money's worth.
- ★ It is clear from the definition that every business transaction involves a transfer and as such consists of two aspects:
 - a) the receiving aspect and
 - b) the giving aspect
- ★ The record of any business transaction will be recording of the two aspects of each transaction is known as 'Double Entry' and the system is called 'Double Entry System'.
- ★ Thus, every transaction affects two accounts and according to Double Entry System entries will be made in both of them – on the debit side in one account and on credit side in the other.
- ★ Hence for every debit there must be a corresponding credit for an equal amount and vice versa. This is known as the 'Principles of Double Entry.'

Advantages of Double Entry System:

- 1) **Scientific System:** Double Entry System records, classifies and summaries business transactions in a systematic manner and, thus produces useful information for decision-makers.
- 2) **Full Information:** Full information can be had about all transactions as the Trader maintains the Ledger with all types of accounts.
- 3) **Assessment of Profit and Loss:** The businessman/ Trader will be able to know correctly whether he had earned profit or sustained loss. It facilitates the Trader to take such steps so as to increase the efficiency of the firm.
- 4) **Know of Debtors:** The Trader will be able to know exactly what amounts are owed by different customers to the firm. If any amount is pending for a long period from any customer, it identifies very easily.
- 5) **Know of Creditors:** The Trader also knows the exact amounts owed by the firm to others and he will be able to arrange prompt payment to obtain cash discount.
- 6) **Assessment of Financial Position:** The Balance Sheet which will help the interested parties to know fully about the financial position of the Firm.
- 7) **Detection of Frauds:** The systematic and scientific recording of business transactions on the basis of this system minimizes the chances of frauds and errors. They can be easily detected by vouching, verification and auditing of accounts.
- 8) **Arithmetical Accuracy:** It provides a check on the arithmetical accuracy with the help of Trial Balance.
- 9) **Comparison of results:** It facilitates the comparison of current year's results with those previous Years. Such comparison helps the owner to manage his business on better lines.
- 10) **Maintenance according to Tax Rules:** Proper maintenance of books will satisfy the Tax authorities and facilitates accurate assessment. In India, Joint Stock Companies should maintain accounts under Double Entry System.

Types Of Accounts and Rule Governing Their Each Account:

★ All business transactions are broadly classified into three categories.

- (i) those relating to persons
- (ii) those relating to property (asset) and
- (iii) those relating to income and expenses

CLASSIFICATION OF ACCOUNTS

Accounts can be classified as

1. **Personal Accounts**

2. **Real Accounts**

3. **Nominal Accounts.**

1. **Personal Accounts :** These accounts deal with persons, whether human beings or artificial persons firms and like companies. For these accounts the rule is

Debit the receiver

Credit the giver.

2. **Real Accounts :** These accounts deal with assets of all kinds. For these accounts the rule is

Debit what comes in

Credit what goes out.

3. **Nominal Accounts :** These accounts deal with items like expenses, losses, incomes and gains (profit) the rule for these accounts is

Debit the expenses and losses

Credit the incomes and gains.

3608 GENERALLY ACCEPTED ACCOUNTING PRINCIPLES

Accounting principles are the guidelines which provide a procedure to be followed in the process of accounting. But it is very difficult to have principles which are universally acceptable. So, we have a set of principles, which are generally accepted by the makers of financial statements. These principles are called as "Generally Accepted Accounting Principles" (GAAP).

Accounting Principles can be categorised as

1. Accounting concepts
2. Accounting Conventions

1) ACCOUNTING CONCEPTS

- a) **Business entity concept** : According to the concept proprietor is different from business and business is different from proprietor. A business organisation is considered to be a separate entity. Accounts of business organisations are prepared in the name of the business not in the name of the proprietors. Infact, a proprietor is considered to be the investor of capital to start run and the business. Thus, the accounts are prepared to present the financial position of the business not the owners.
- b) **Going concern concept** : It is assumed that a business organisation will continue to operate for a considerably long period of time. This assumption is very important because some times money is spent by the business for a future benefit.
- c) **Money measurement concept** : Business organisations are engaged in economic activity. It means they deal with money. Therefore, in the process of accounting only events which can be expressed and measured in terms of money are recorded.
- d) **Cost concept** : According to this concept, all the transactions are recorded at their cost. It means all the transactions are recorded at the price at which the exchange was completed. The market value or any other value of transaction is not considered. It is therefore called as "historical cost concept".
- e) **Accounting period concept** : Business organisations are believed to have continued existence. But, the performance of the organisation should be assessed regularly, over a period of time. Normally one year is taken to assess the financial performance of the business.
- f) **Dual aspect concept** : Dual means two. Dual aspect concept implies that every transaction, completed in any organisation, will have two aspects. These aspects are giving and receiving.
- g) **Matching concept** : This concept tries to match the revenue and expenditure of an organisation pertaining to a particular period of time i.e., 1 year. This will help the users of accounting to understand the functioning of the business in a clear manner.

- h) Realisation concept :** According to this concept revenue is deemed to be realised when it is earned. "Earning" does not mean realising cash. It means any earning process which is complete and any transaction where process is complete but money is not realised.

2) **ACCOUNTING CONVENTIONS**

- a) Convention of materiality :** According to this convention, accounting should consist of all the material facts. Material facts are those which can have an impact on the financial statements of the organisation. Immaterial and irrelevant items should be excluded or merged with other items.
- b) Convention of disclosure :** The user of the financial statements should be made available with all the material facts relating to the business. No item, of material nature, should be kept secret.
- c) Convention of consistency :** This convention of consistency implies that a particular accounting system or procedure or method or treatment followed in one year should be followed in the future also.
- d) Convention of Conservation :** According to this convention all incomes or profits if anticipated should not be taken into account. But any anticipated losses should be considered while preparing the accounts of a business.

TRIAL BALANCE OF as on

Name of the account	Debit (Rs.)	Credit (Rs.)
Capial Account		X X X
Drawsings Account	X X X	
Assets Account	X X X	
Liabilities Account		X X X
Purchases Account	X X X	
Purchase Return Account		X X X
Sales Account		X X X
Sales Return Account	X X X	
Reserves & Provisions Account		X X X
Expenses & Losses Account	X X X	
Incomes & gains Account		X X X
Outstanding Expenses Account		X X X
Prepaid Expenses Account	X X X	
(Unexpired Expenses Account)		
Income Receivable Account	X X X	
(Accrued Income Account)		
Income Recieved in Advance		X X X
Total	X X X X	X X X X

FINAL ACCOUNTS

Final accounts are prepared by an organisation at the end of the financial year to know the operational efficiency and financial position of the business financial accounts for a trading firm refer to

- 1) Trading and profit and Loss Account
- 2) Balance Sheet

Trial balance is the basis for preparation of final accounts.

TRADING ACCOUNT

This account is prepared to ascertain the profit or Loss earned by the business from buying and selling of goods manufactured during a particular period of time. Normally, the period is a financial year. The profit earned in this account is called as "Gross Profit" and the Profit or Loss earned in this account is transferred to profit and Loss account.

PROFORMA TRADING ACCOUNT

TRADING ACCOUNT			
Dr.			Cr.
Particulars	Amount Rs.	Particulars	Amount Rs.
To opening stock	X X X	By sales	X X X
To purchases X X X		(-) sales returns X X	X X X
(-) Purchase returns X X	X X X	By closing stock	X X X
To carriage cartage freight inwards	X X X	By Gross loss (if any) c/f	X X X
To Wages	X X X		
To manufacturing expenses	X X X		
To Fuel and power	X X X		
	X X X		
To Import duty	X X X		
To Factory Expenses	X X X		
To Gross Profit c/f	X X X		
(Transferred to P&L a/c)	X X X		
			X X X

PROFIT AND LOSS ACCOUNT

Dr.

Cr.

Particulars	Amount	Particulars	Amount
To Gross Loss (if any) b/f	X X X	By Gross Profit b/f	X X X
To printing & stationery	X X X	By Rent received	X X X
To salaries	X X X	By Commission received	X X X
To lighting & telephone charges	X X X	By Interest received	X X X
To commission paid	X X X	By other incomes	X X X
To Depreciation on assets	X X X	By Net Loss, (if any)	X X X
To Discount allowed	X X X	(Transferred to capital	X X X
To postage & Telegrams	X X X	account)	
To carriage Cartage/Freight outwards	X X X		
To Rent rates & taxes	X X X		
To Insurance	X X X		
To Advertisements	X X X		
To general expenses	X X X		
To Net Profit	X X X		
(Transferred to capital account)			X X X

FORMAT OF BALANCE SHEET

BALANCE SHEET

Liabilities		Amount	Assets	Amount
<u>Current Liabilities</u>			<u>Current Assets</u>	
Bills payable		X X X	Cash in hand	X X X
Bank overdraft		X X X	Cash at bank	X X X
Sundry Creditors		X X X	Bills receivable	X X X
Outstanding Expenses		X X X		
<u>Long term Liabilities</u>			Sundry debtors	X X X
			Prepaid Expenses	X X X
			Closing Stock	X X X
<u>Capital</u>			<u>Fixed Assets</u>	
Opening Balance	X X X		Investments	X X X
			Motor vehicles	X X X
(+) Net profit	X X		Patents	X X X
(+) Interest on capital	XX		Trade marks	X X X
(-) Drawings	X X		Furniture	X X X
(-) Interest on Drawings	X X		Land & buildings	X X X
(-) Net Loss	X X	X X X	Plant and machinery	X X X
Long term loan		X X X		X X X
		X X X		

Capital budgeting Techniques:

The capital budgeting appraisal methods are techniques of evaluation of investment proposal will help the company to decide upon the desirability of an investment proposal depending upon their; relative income generating capacity and rank them in order of their desirability. These methods provide the company a set of norms on the basis of which either it has to accept or reject the investment proposal. The most widely accepted techniques used in estimating the cost-returns of investment projects can be grouped under two categories.

1. Traditional methods
2. Discounted Cash flow methods

1. Traditional methods

These methods are based on the principles to determine the desirability of an investment project on the basis of its useful life and expected returns. These methods depend upon the accounting information available from the books of accounts of the company. These will not take into account the concept of 'time value of money', which is a significant factor to determine the desirability of a project in terms of present value.

A. Pay-back period method: It is the most popular and widely recognized traditional method of evaluating the investment proposals. It can be defined, as 'the number of years required to recover the original cash out lay invested in a project'.

According to Weston & Brigham, "The pay back period is the number of years it takes the firm to recover its original investment by net returns before depreciation, but after taxes".

According to James. C. Vanhorne, "The payback period is the number of years required to recover initial cash investment.

The pay back period is also called payout or payoff period. This period is calculated by dividing the cost of the project by the annual earnings after tax but before depreciation under this method the projects are ranked on the basis of the length of the payback period. A project with the shortest payback period will be given the highest rank and taken as the best investment. The shorter the payback period, the less risky the investment is the formula for payback period is

$$\text{Pay-back period} = \frac{\text{Cash outlay (or) original cost of project}}{\text{Annual cash inflow}}$$

Merits:

1. It is one of the earliest methods
2. It is simple to understand and to compute.
3. It dose not involve any cost for computation of the payback period
4. It is one of the widely used methods in small scale industry sector
5. It can be computed on the basis of accounting information available from the books.

Demerits:

1. This method fails to take into account the cash flows received by the company after the pay back period.
2. It doesn't take into account the interest factor involved in an investment outlay.
3. It doesn't take into account the interest factor involved in an investment outlay.
4. It is not consistent with the objective of maximizing the market value of the company's share.
5. It fails to consider the pattern of cash inflows i. e., the magnitude and timing of cash in flows.

B. Accounting (or) Average rate of return method (ARR):

It is an accounting method, which uses the accounting information repeated by the financial statements to measure the probability of an investment proposal. It can be determine by dividing the average income after taxes by the average investment i.e., the average book value after depreciation.

According to 'Soloman', accounting rate of return on an investment can be calculated as the ratio of accounting net income to the initial investment, i.e.,

$$\text{ARR} = \frac{\text{Average net income after taxes}}{\text{Average Investment}} \times 100$$

$$\begin{aligned} \text{Average net income after taxes} &= \frac{\text{Total Income after Taxes}}{\text{No. Of Years}} \\ \text{Average investment} &= \frac{\text{Total Investment}}{2} \end{aligned}$$

If working capital of Scrap is given

$$\text{Average investment} = \frac{\text{Cost-Scrap} + \text{working capital} + \text{Scrap.}}{2}$$

On the basis of this method, the company can select all those projects who's ARR is higher than the minimum rate established by the company. It can reject the projects with an ARR lower than the expected rate of return. This method can also help the management to rank the proposal on the basis of ARR. A highest rank will be given to a project with highest ARR, where as a lowest rank to a project with lowest ARR.

Merits:

It is very simple to understand and calculate.

1. It can be readily computed with the help of the available accounting data.
2. It uses the entire stream of earning to calculate the ARR.

Demerits:

1. It is not based on cash flows generated by a project.
2. This method does not consider the objective of wealth maximization
3. IT ignores the length of the projects useful life.
4. It does not take into account the fact that the profits can be re-invested.

II: Discounted cash flow methods:

The traditional method does not take into consideration the time value of money. They give equal weight age to the present and future flow of incomes. The DCF methods are based on the concept that a rupee earned today is more worth than a rupee earned tomorrow. These methods take into consideration the profitability and also time value of money.

A. Net present value method (NPV)

The NPV takes into consideration the time value of money. The cash flows of different years and valued differently and made comparable in terms of present values for this the net cash inflows of various period are discounted using required rate of return which is predetermined.

According to Ezra Solomon, "It is a present value of future returns, discounted at the required rate of return minus the present value of the cost of the investment."

NPV is the difference between the present value of cash inflows of a project and the initial cost of the project.

According the NPV technique, only one project will be selected whose NPV is positive or above zero. If a project(s) NPV is less than 'Zero'. It gives negative NPV hence. It must be rejected. If there are more than one project with positive NPV's the project is selected whose NPV is the highest.

The formula for NPV is

NPV= Present value of cash inflows – Cash outflow.

PV VALUES = $1/(1+r)^n$

Where r= discount rate %

. n = no of years

Merits:

1. It recognizes the time value of money.
2. It is based on the entire cash flows generated during the useful life of the asset.
3. It is consistent with the objective of maximization of wealth of the owners.
4. The ranking of projects is independent of the discount rate used for determining the present value.

Demerits:

1. It is different to understand and use.
2. The NPV is calculated by using the cost of capital as a discount rate. But the concept of cost of capital. If self is difficult to understood and determine.
3. It does not give solutions when the comparable projects are involved in different amounts of

B. Internal Rate of Return Method (IRR)

The IRR for an investment proposal is that discount rate which equates the present value of cash inflows with the present value of cash out flows of an investment. The IRR is also known as cutoff or hurdle rate. It is usually the concern's cost of capital.

According to Weston and Brigham "The internal rate is the interest rate that equates the present value of the expected future receipts to the cost of the investment outlay.

When compared the IRR with the required rate of return (RRR), if the IRR is more than RRR then the project is accepted else rejected. In case of more than one project with IRR more than RRR, the one, which gives the highest IRR, is selected.

**** Calculate two NPVs at different cost of capital.

$$\text{IRR} = L + \frac{\text{PVL} - \text{Cashoutflow}}{\text{PVL} - \text{PVH}} \times H - L$$

L- Lower discount rate

P1 - Present value of cash inflows at lower rate.

P2 - Present value of cash inflows at higher rate.

Q- Actual investment

D- Difference in Discount rates.

Merits:

1. It consider the time value of money
2. It takes into account the cash flows over the entire useful life of the asset.
3. It has a psychological appeal to the user because when the highest rate of return projects are selected, it satisfies the investors in terms of the rate of return on capital
4. It always suggests accepting to projects with maximum rate of return.
5. It is in conformity with the firm's objective of maximum owner's welfare.

Demerits:

1. It is very difficult to understand and use.
2. It involves a very complicated computational work.
3. It may not give unique answer in all situations.

C. Probability Index Method (PI)

The method is also called benefit cost ratio. This method is obtained with a slight modification of the NPV method. In case of NPV the present value of cash out flows are profitability index (PI), the

present value of cash inflows are divide by the present value of cash out flows, while NPV is a absolute measure, the PI is a relative measure.

It the PI is more than one (>1), the proposal is accepted else rejected. If there are more than one investment proposal with the more than one PI the one with the highest PI will be selected. This method is more useful incase of projects with different cash outlays cash outlays and hence is superior to the NPV method.

The formula for PI is

$$\text{Probability index} = \frac{\text{Present Value of Future Cash Inflow}}{\text{Investment}}$$

Merits:

1. It requires less computational work then IRR method
2. It helps to accept / reject investment proposal on the basis of value of the index.
3. It is useful to rank the proposals on the basis of the highest/lowest value of the index.
4. It is useful to tank the proposals on the basis of the highest/lowest value of the index.

Demerits:

1. It is some what difficult to understand
2. Some people may feel no limitation for index number due to several limitation involved in their competitions
3. It is very difficult to understand the analytical part of the decision on the basis of probability index.